Zero Crossings on the Accuracy-Smoothness Frontier: Maximal Monotone and Minimal Inflection Non Stationary Predictors

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Abstract

The Smooth Sign Accuracy (SSA) framework resolves the smoothness–accuracy dilemma for stationary predictors by balancing mean squared error (MSE) against the frequency of sign changes. We generalize it to non-stationary integrated processes (I-SSA), deriving MSE-optimal predictors constrained by the expected duration between turning points (I(1)) or inflection points (I(2)). The resulting maximal-monotone and minimal-inflection predictors, filters, and smoothers customize classic benchmarks along a new efficient Accuracy–Smoothness (AS) frontier. Tracking the identity (the original data) along this frontier defines the *I-SSA trend*, a novel smoothing concept that, unlike classical filters, imposes no artificial structure on the data-generating process and is directly interpretable through explicit control of turning- or inflection-point frequency. The optimization is a quadratically constrained quadratic program (QCQP) that reduces to a Generalized Trust Region Subproblem (GTRS) with hidden convexity. On its feasible, invertible region the problem is convex with a well-defined global optimum, and an exact closed-form expression delivers the efficient AS frontier. Under fairly general assumptions, finite-sample solutions converge to an asymptotic I-SSA predictor, devoid of integration boundary perturbations. The framework unifies mean squared error on non-stationary levels, and sign accuracy, zero-crossing frequency, and target correlation on stationary differences.

Keywords: Time series, sign accuracy, mean-squared error, forecasting dilemma, zero crossing rate, signal extraction, non-stationarity, efficient accuracy smoothness frontier (62M10).

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1 Introduction

Forecasting requires balancing conflicting objectives. The SSA framework (Wildi, 2024, 2026a) targets accuracy (estimating future levels) and smoothness (suppressing spurious noise). We propose I-SSA, extending SSA to non-stationary integrated series and letting users balance accuracy (A) and smoothness (S) along a novel, efficient AS-frontier. We focus on univariate, homoscedastic series (the multivariate stationary case is treated in Wildi, 2026b). I-SSA overcomes the limitations of the frequency-domain forecast trilemma (McElroy and Wildi, 2019), offering better interpretability, numerical efficiency, and robustness against overfitting.

Traditional smoothing methods include curvature-penalized approaches (Whittaker, 1923; Henderson, 1924; Hodrick and Prescott, 1997)—despite recent critiques (Hamilton, 2018; Phillips and Shi, 2021)—splines (Eilers, Marx, and Durban, 2015; Wood, 2017), trend extraction (Harvey, 1989; Durbin and Koopman, 2012), and spectral filtering (Baxter and King, 1999; Pollock, 2000; Percival and Walden, 2000). Since our approach emphasizes turning and inflection points (TP/IP), a natural benchmark is the HP filter, a special case of Whittaker–Henderson smoothing, still widely used in business-cycle analysis and crisis prediction (Drehmann and Yetman, 2021); we complement the analysis with the boosted HP filter (Phillips and Shi, 2021, 2024). Broader comparisons are available in our R package.

While the HP filter minimizes curvature, SSA defines smoothness via the expected duration between sign changes (zero crossings, ZC) of a stationary, zero-mean predictor, building on Rice (1944). I-SSA extends this to integrated processes by targeting ZCs in stationary differences while minimizing MSE at the non-stationary levels. Economically, sign changes in growth rates mark transitions between expansions and contractions, so capturing business cycles requires sufficiently long intervals between crossings. This mean duration, the holding time (HT) (Wildi, 2024), is explicitly regulated in I-SSA for the growth (I(1)) or acceleration (I(2)) of the predictor via the rate of TPs or IPs.

I-SSA defines a novel Accuracy-Smoothness (AS) frontier: no other linear predictor of the same, or shorter length can improve both MSE on non-stationary levels (accuracy) and ZC rate in stationary differences (smoothness). Within this family, I-SSA is characterized by the fewest growth sign-changes, which justifies the terms maximal monotone (I(1)) and minimal inflection (I(2)).[1] The classic MSE predictor is one point on this frontier, whereas I-SSA covers it in full. We also give a closed-form link between HT and MSE along the frontier, extending the stationary AS dilemma (Wildi, 2024, 2026a,b) to integrated processes.

Extending SSA to the non-stationary setting raises technical caveats. The cointegration constraint between the acausal target and the causal predictor, which ensures finite MSE on levels, introduces asymmetries in the Toeplitz structure of the system matrices. Permanent 'unit-root' memory prevents convergence to circulants, so the system matrices do not commute and a fixed Fourier eigenbasis is unavailable. We therefore use a generalized eigenvalue decomposition of a matrix pencil, as in cointegration (Johansen, 1988, 1991) and canonical correlation analysis (Box and Tiao, 1977), but with a structure originating in the optimization literature on the generalized trust-region subproblem (GTRS; Moré, 1993; Adachi et al., 2017) because the Hessian is generally indefinite. The pencil separately encodes the non-stationary level and stationary autocorrelation operators, which the generalized decomposition diagonalizes simultaneously. This yields a closed-form, MSE-optimal predictor constrained by the expected duration between turning points, to our knowledge not previously explored for integrated processes. Although the formulation matches the stationary case, the geometry differs fundamentally: the generalized eigenvectors are non-orthogonal and depend on the HT, and the acausal target and causal predictor rely on different spectral bases, leading to a counterfactual spectral decomposition (details below).

Formally, the optimization is a generalized trust-region subproblem (GTRS) with hidden convexity. If the classic MSE predictor is known, I-SSA reduces to a convolution with it that modifies its ZC rate while preserving MSE optimality; we call this transformation customization. For this

[1] The method also yields minimal monotone and maximal inflection predictors for a given MSE, though such extreme 'un-smoothing' is typically irrelevant in prediction; see Wildi (2026a,b).

problem, the region of invertibility and strict monotonicity reduce prediction to a one-dimensional search with certification of the global optimum. Under fairly general assumptions, finite-length solutions converge to a well-structured infinite-length I-SSA problem, free of finite-length boundary perturbations.

I-SSA customization means the projection of existing benchmark predictors onto the non-stationary AS frontier. In the applications, we customize the HP filter and its boosted variant, introducing the maximal monotone HP trend nowcast, obtained by targeting the acausal two-sided (boosted) HP filter in I-SSA. This reconciles growth sign analysis, in differences, with optimal level tracking. With the cursor on the frontier set to replicate the benchmark's TP rate, the I-SSA trend achieves lower MSE, notably at TPs, with faster response (smaller retardation) and no over- or undershooting. All examples are replicable with our open-source R package (`https://github.com/wiaidp/R-package-SSA-Predictor`), which also extends customization to Hamilton's regression filter (Hamilton, 2018), the Baxter–King filter (Baxter and King, 1999), and a refined Beveridge–Nelson decomposition (Kamber, Morley, and Wong, 2024).

Replacing the two-sided HP target by the identity, i.e., the original data, defines the *I-SSA trend*: minimal MSE subject to maximal monotonicity (I(1)) or minimal inflection (I(2)). This novel smoothing principle imposes minimal structural assumptions on the data generating process (DGP), consistent with the typical spectral shape of economic time series (Granger, 1966), and prescribes no functional form (polynomial, spline, minimum curvature). TPs or IPs emerge from the interaction between the 'neutral' (amorphous-memory) HT constraint and optimal tracking, which supports efficiency and interpretability. Relative to the classic Whittaker–Henderson smoother, the I-SSA trend yields a lower nowcast MSE at directional changes, confirming the findings for the HP customization. Moreover, the concurrent I-SSA trend distributes TPs more evenly and gives clearer signals, avoiding clusters of spurious TPs around directional changes.

The remainder of the paper is organized as follows. Section 2 reviews the SSA criterion (Wildi, 2024, 2026a,b); Section 3 extends it to integrated processes through I-SSA; Section 4 applies customization to business cycle nowcasting and the I-SSA trend to data smoothing; Section 5 concludes.

2 SSA

The following summarizes the stationary Smooth Sign Accuracy framework of Wildi (2024, 2026a,b), which provides the foundation for I-SSA.

Consider a zero-mean stationary series x_t (non-zero-mean processes are centered) $x_t = \sum_{i=0}^{\infty} \xi_i \epsilon_{t-i}$, with $\xi_0 = 1$. The one-sided sequence $\boldsymbol{\xi}_\infty := (\xi_0, \xi_1, ...)'$ is square summable and corresponds to the weights in the (purely non-deterministic) Wold decomposition of x_t, as detailed by Brockwell and Davis (1993). Let $z_t = \sum_{k=-\infty}^{\infty} \gamma_k x_{t-k}$ be another stationary process and assume we want to predict $z_{t+\delta}$, the prediction target, with $\delta \in \mathbb{Z}$. We construct a length-L causal predictor $y_t = \sum_{k=0}^{L-1} b_k x_{t-k}$ for $z_{t+\delta}$, termed forecasting, nowcasting, or backcasting as $\delta > 0$, $= 0$, or < 0.[2]

For example, with $x_t = \epsilon_t$, $\gamma_0 = 1$, $\gamma_1 = 0.5$, $z_t = \epsilon_t + 0.5\epsilon_{t-1}$ is an MA(1); one-step forecasting ($\delta = 1$) gives the MSE predictor $y_t = 0.5\epsilon_t$, i.e., $b_0 = 0.5$, $b_k = 0$ otherwise.

The MSE predictor tracks the level optimally but can be 'noisy': here $y_t = 0.5\epsilon_t$ is WN while z_{t+1} is more persistent. Similarly, in real-time signal extraction the one-sided concurrent filter is markedly noisier than its acausal, two-sided counterpart (see examples below). To address this, Wildi (2024) proposes to control the HT, the mean duration between consecutive zero-crossings, of the predictor.

Specifically, let $\boldsymbol{\Xi}$ denote the $L \times L$ matrix with i-th row $\boldsymbol{\Xi}_{i\cdot} := (\xi_{i-1}, \xi_{i-2}, ..., \xi_0, \mathbf{0}_{L-i})$, $i = 1, ..., L$, where $\mathbf{0}_{L-i}$ is a zero vector of length $L - i$. We define $\mathbf{x}_t := (x_t, ..., x_{t-(L-1)})'$ and $\boldsymbol{\epsilon}_t := (\epsilon_t, ..., \epsilon_{t-(L-1)})'$. Additionally, we define $\mathbf{b}_\epsilon := \boldsymbol{\Xi}\mathbf{b}_x$. Consequently, we obtain:

$$y_t = \mathbf{b}_x' \mathbf{x}_t \approx (\boldsymbol{\Xi}\mathbf{b}_x)' \boldsymbol{\epsilon}_t = \mathbf{b}_\epsilon' \boldsymbol{\epsilon}_t,$$

[2] Our target specification also accommodates signal extraction, where γ_k are coefficients of a two-sided (possibly bi-infinite) filter, see our examples.

where the approximation via the finite MA inversion of x_t holds if the filter coefficients decay to zero sufficiently rapidly or, equivalently, if L is sufficiently large (exact results can be derived but are omitted here). Wildi (2024,2026a) proposes the following SSA criterion

$$\min_{\mathbf{b}_\epsilon} E[(z_{t+\delta} - y_t)^2] \tag{1}$$
$$\mathbf{b}_\epsilon' \mathbf{M} \mathbf{b}_\epsilon - \rho_1 \mathbf{b}_\epsilon' \mathbf{b}_\epsilon = 0, \tag{2}$$

where the MA approximation holds at the stationary point of the Lagrangian (see details below). The $L \times L$ autocovariance generating matrix $\mathbf{M}$,

$$\mathbf{M} = \begin{pmatrix} 0 & 0.5 & 0 & 0 & 0 & ... & 0 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 & 0 & ... & 0 & 0 & 0 \\ ... & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & ... & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 0 & 0 & ... & 0 & 0.5 & 0 \end{pmatrix},$$

satisfies $\mathbf{b}_\epsilon{}' \mathbf{M} \mathbf{b}_\epsilon = \sum_{k=1}^{L-1} b_{\epsilon,k-1} b_{\epsilon,k}$, the first-order autocovariance of the predictor, assuming standardized white noise. The constraint (2) imposes a first-order autocorrelation (ACF(1)) of ρ_1 and is feasible if $|\rho_1| < \rho_{\max}(L)$, the largest eigenvalue of $\mathbf{M}$ determining the Rayleigh quotient. Since $|\rho_1| < \rho_{\max}(L)$, the matrix $\mathbf{M} - \rho_1 \mathbf{I}$ is indefinite, with both positive and negative eigenvalues, which gives the SSA problem its GTRS structure.

Let $ht(y|\mathbf{b}_\epsilon)$ denote the (stationarity-invariant) expected duration between consecutive zero-crossings of y_t, the HT.[3] Assuming Gaussianity,

$$ht(y|\mathbf{b}_\epsilon) = \frac{\pi}{\arccos(\rho(y))}, \tag{3}$$

linking HT and ACF(1) (Wildi, 2024); this relation is fairly robust to non-Gaussianity (central limit theorem, see Wildi, 2026b). Since arccos is monotonic on $]-1, 1[$, the criterion becomes

$$\left.\begin{array}{c} \min_{\mathbf{b}_\epsilon} E[(z_{t+\delta} - y_t)^2] \\ ht(y|\mathbf{b}_\epsilon) = ht_1 \end{array}\right\}.$$

Specifying $ht_1 > ht_{MSE}$ ($ht_1 < ht_{MSE}$), the HT of the MSE predictor, yields a smoother (unsmoother) predictor with fewer (more) ZCs. The resulting SSA predictor minimizes MSE subject to the ZC rate; dually, it minimizes the ZC rate for given MSE (efficient frontier). The MSE predictor is recovered by setting $ht_1 = ht_{MSE}$. Wildi (2024,2026b) proposes variants of the SSA criterion in which the MSE objective (1) is replaced by either the target correlation, i.e., the correlation between the predictor y_t and the target $z_{t+\delta}$, or the sign accuracy, i.e., the probability that predictor and target have the same sign (assuming Gaussianity). Since all variants are equivalent, SSA unifies MSE, target correlation, sign accuracy, and the rate of zero-crossings within a single interpretable framework.

3 I-SSA

Here $\tilde{x}_t$, $\tilde{z}_t = \sum_{|k|<\infty} \gamma_k \tilde{x}_{t-k}$, and $y_t = \sum_{k=0}^{L-1} b_{\tilde{x}k} \tilde{x}_{t-k}$ are all assumed non-stationary I(d), with $d > 0$, so HT and zero-crossing rate, autocorrelation, target correlation or sign accuracy become ill-defined; we now show how to resolve this.

3.1 Concept

Let $\Delta(B)\tilde{x}_t = (1-B)^d \tilde{x}_t =: x_t$ be stationary and invertible, with Wold weights ξ_k, $k \geq 0$. The MSE predictor is $y_{t,MSE} = \hat{\boldsymbol{\gamma}}'_{MSE} \tilde{\mathbf{x}}_t$, with weights $(\hat{\gamma}_{\tilde{x}\delta k})_{k=0,...,L-1}$ and $\tilde{\mathbf{x}}_t := (\tilde{x}_t, ..., \tilde{x}_{t-(L-1)})'$,

[3] Crossings satisfy $t_{i-1} < t_i$, $t_1 \geq L$, $y_{t_i-1} y_{t_i} < 0$, and $y_{t-1} y_t > 0$ for $t_{i-1} < t < t_i$.

valid under general assumptions (McElroy and Wildi, 2016; McElroy and Livsey, 2022; McElroy, 2022). We henceforth assume the MSE predictor $(\hat{\gamma}_{\tilde{x}\delta k})_{k=0,...,L-1}$ to be known.

The MSE principle guarantees stationarity of $e_t := \tilde{z}_{t+\delta} - y_{t,MSE}$, so $\tilde{z}_{t+\delta}$ and $y_{t,MSE}$ are cointegrated with vector $(1,-1)$. To eliminate the unit root(s) of $\tilde{x}_t$, the error filter $\gamma_{k+\delta} - \hat{\gamma}_{\tilde{x}\delta k}$ must satisfy

$$\sum_{k=-\infty}^{\infty} (\gamma_{k+\delta} - \hat{\gamma}_{\tilde{x}\delta k}) k^j = 0, \quad j = 0, ..., d-1, \tag{4}$$

with $\hat{\gamma}_{\tilde{x}\delta k} = 0$ for $k \notin \{0, ..., L-1\}$, (McElroy and Wildi, 2016), see below for details.

We focus on $d = 1$, so $\tilde{x}_t, \tilde{z}_t \sim I(1)$; the extension to $d > 1$ is analogous (Appendix (B)). We assume throughout that $\sum_{k=-\infty}^{\infty} \gamma_{k+\delta} \neq 0$, with continuous transfer function at zero frequency, so that the filter does not cancel the unit root, i.e., $\tilde{z}_t$ is non-stationary I(1) (if $\tilde{z}_t$ is stationary, the problem can be re-framed in SSA through suitable transformations). Define the $L \times L$ summation and differencing matrices

$$\mathbf{\Sigma} = \begin{pmatrix} 1 & 0 & 0 & ... & 0 \\ 1 & 1 & 0 & ... & 0 \\ ... & & & & \\ 1 & 1 & 1 & ... & 1 \end{pmatrix} \text{ and } \mathbf{\Delta} := \mathbf{\Sigma}^{-1} = \begin{pmatrix} 1 & 0 & 0 & ... & 0 & 0 \\ -1 & 1 & 0 & ... & 0 & 0 \\ & & & & & \\ 0 & 0 & 0 & ... & -1 & 1 \end{pmatrix}.$$

For a predictor y_t with weights $\mathbf{b}$, define $e_{y,t} := y_{t,MSE} - y_t$, replacing target $\tilde{z}_{t+\delta}$ with the MSE benchmark $y_{t,MSE}$—justified by the law of iterated projections: the MSE predictor serves as an equivalent I-SSA target, and I-SSA is a constrained predictor of $\tilde{z}_{t+\delta}$ or a smoother of the MSE predictor. The I(1)-SSA filter $\mathbf{b}_{\tilde{x}} = (b_{\tilde{x}0}, ..., b_{\tilde{x}L-1})'$ minimizes $E[e_{y,t}^2]$ subject to the HT constraint (detailed below); without constraint it reduces to the MSE design ($\mathbf{b}_{\tilde{x}} = \hat{\boldsymbol{\gamma}}_{MSE}$ and $e_{y,t} = 0$). The added cointegration constraint

$$\sum_{k=0}^{L-1} b_{\tilde{x}k} = \sum_{k=0}^{L-1} \hat{\gamma}_{\tilde{x}\delta k}$$

ensures $e_{SSA,t}$ is stationary:

$$e_{SSA,t} = (\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\tilde{\mathbf{x}}_t = (\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\mathbf{\Sigma}'\mathbf{\Delta}'\tilde{\mathbf{x}}_t = (\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\mathbf{\Sigma}'\mathbf{x}_t, \tag{5}$$

the last equality holding because the final entry of $(\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\mathbf{\Sigma}'$ vanishes by the cointegration constraint, allowing the non-stationary $\tilde{x}_{t-(L-1)}$ in $\mathbf{\Delta}'\tilde{\mathbf{x}}_t$ to be replaced by the stationary $x_{t-(L-1)} = \tilde{x}_{t-(L-1)} - \tilde{x}_{t-L}$ (or any number). Assuming L large enough, the finite MA approximation[4]

$$(\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\mathbf{\Sigma}'\mathbf{x}_t \approx (\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\mathbf{\Sigma}'\mathbf{\Xi}'\boldsymbol{\epsilon}_t$$

gives

$$\begin{aligned} e_{SSA,t} &= y_{t,MSE} - y_t = (\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\mathbf{\Sigma}'\mathbf{x}_t \approx (\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\mathbf{\Sigma}'\mathbf{\Xi}'\boldsymbol{\epsilon}_t \\ &= \hat{\boldsymbol{\gamma}}'_{MSE}\mathbf{\Sigma}'\mathbf{\Xi}'\boldsymbol{\epsilon}_t - \mathbf{b}'_{\epsilon}\mathbf{\Sigma}'\boldsymbol{\epsilon}_t, \end{aligned} \tag{6}$$

where $\mathbf{b}_{\epsilon} := \mathbf{\Xi}\mathbf{b}_{\tilde{x}}$ inherits the cointegration constraint from $\mathbf{b}_{\tilde{x}}$ (see below), and the last equality uses commutativity of $\mathbf{\Sigma}$ and $\mathbf{\Xi}$ under convolution (lower-triangular Toeplitz matrices). In (6), $y_{t,MSE}$ and y_t are non-stationary cointegrated predictors of $\tilde{z}_{t+\delta}$, while the right-hand processes $\hat{\boldsymbol{\gamma}}'_{MSE}\mathbf{\Sigma}'\mathbf{\Xi}'\boldsymbol{\epsilon}_t$ and $\mathbf{b}'_{\epsilon}\mathbf{\Sigma}'\boldsymbol{\epsilon}_t$ are stationary and no longer predictors—yet their cross-sectional differences remain (virtually) identical, which is key to optimization.

The stationary MA representations in (6) thus yield a valid target correlation for the objective, while an HT constraint can be based on $\mathbf{b}'_{\epsilon}\boldsymbol{\epsilon}_t = \mathbf{b}'_{\tilde{x}}\mathbf{\Xi}'\boldsymbol{\epsilon}_t \approx y_t - y_{t-1}$, controlling zero-crossings of

[4]Theorem 2 in the main theoretical results, Section 3.3 below, shows that filter weights decay sufficiently fast for the MA approximation to be valid under fairly general assumptions on the target.

the stationary first difference $y_t - y_{t-1}$; a length constraint on its variance is likewise feasible. This yields a conceptual I(1)-SSA criterion:

$$\left.\begin{array}{c} \min_{\mathbf{b}_\epsilon} (\boldsymbol{\gamma}_{\tilde{\Xi}\delta} - \boldsymbol{\Sigma}\mathbf{b}_\epsilon)'(\boldsymbol{\gamma}_{\tilde{\Xi}\delta} - \boldsymbol{\Sigma}\mathbf{b}_\epsilon) \\ \mathbf{b}_\epsilon'\mathbf{M}\mathbf{b}_\epsilon = \rho_1 \mathbf{b}_\epsilon'\mathbf{b}_\epsilon \end{array}\right\}, \tag{7}$$

where $\tilde{\boldsymbol{\Xi}} := \boldsymbol{\Xi}\boldsymbol{\Sigma} = \boldsymbol{\Sigma}\boldsymbol{\Xi}$ and $\boldsymbol{\gamma}_{\tilde{\Xi}\delta} := \tilde{\boldsymbol{\Xi}}\hat{\boldsymbol{\gamma}}_{MSE}$. The criterion presupposes the cointegration constraint 'ex nihilo' and is thus incomplete, but remains useful for interpretability. The target derivative $-2\boldsymbol{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}\delta} + 2\boldsymbol{\Sigma}'\boldsymbol{\Sigma}\mathbf{b}_\epsilon$ gives the Lagrangian normal equation $\left(-\tilde{\lambda}\left(\mathbf{M} - \rho_1\mathbf{I}\right) + \boldsymbol{\Sigma}'\boldsymbol{\Sigma}\right)\mathbf{b}_\epsilon = \boldsymbol{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}\delta}$, hence $\mathbf{b}_\epsilon = F\tilde{\boldsymbol{\mathcal{V}}}^{-1}\boldsymbol{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}\delta}$ with $\tilde{\boldsymbol{\mathcal{V}}} := \mathbf{M} - \rho_1\mathbf{I} - 1/\tilde{\lambda}\boldsymbol{\Sigma}'\boldsymbol{\Sigma}$ and $F := -1/\tilde{\lambda}$. Wildi (2026a) proposes a variant of I-SSA based on the target correlation of the stationary processes on the right of (6); both criteria are equivalent.

This expression extends the SSA solution, in which the integration operator $\boldsymbol{\Sigma}$ is replaced by $\mathbf{I}$, so that the 'reduced' Hessian $\tilde{\boldsymbol{\mathcal{V}}} := \mathbf{M} - \rho_1\mathbf{I} - 1/\tilde{\lambda}\mathbf{I}$ admits a simple decomposition into the Fourier eigenvectors of $\mathbf{M}$, i.e., a fixed geometry. Its presence in I-SSA is the main technical difficulty: $\tilde{\boldsymbol{\mathcal{V}}}$ 'grows' unboundedly, and the classic eigenvalue decomposition now depends on both ρ_1 and $\tilde{\lambda}$, because $\mathbf{M}$ and $\boldsymbol{\Sigma}'\boldsymbol{\Sigma}$ do not commute and cannot be diagonalized simultaneously. The resulting classic geometry is parameter-dependent and unsuited to our purpose.

For a (nearly) Gaussian process y_t, $ht_1 := \pi/\arccos(\rho_1)$ defines the HT of the first differences $y_t - y_{t-1}$, assuming zero mean (if a linear drift is present, the constraint instead addresses crossover points around the trend). Dually, y_t is maximally monotone if sign changes of $y_t - y_{t-1}$ are minimized for a given tracking accuracy (e.g., MSE, detailed below). We next formally integrate the cointegration constraint, imposing additional structure and asymmetry on the problem.

3.2 Solution

Let $\Gamma(0) := \sum_{k=0}^{L-1} \hat{\gamma}_{\tilde{x}\delta k} \neq 0$ be the zero-frequency transfer function of the MSE filter, assuming continuity at zero (McElroy and Wildi, 2016). So the I(1)-SSA cointegration constraint reads $\sum_{k=0}^{L-1} b_{\tilde{x}k} = \Gamma(0) \neq 0$, or in vector form

$$\mathbf{b}_{\tilde{x}} = \Gamma(0)\mathbf{e}_1 + \mathbf{B}\tilde{\mathbf{b}},$$

where $\mathbf{e}_1 = (1, 0, ..., 0)'$ $(L \times 1)$, $\tilde{\mathbf{b}} = (\tilde{b}_1, ..., \tilde{b}_{L-1})'$ $((L-1) \times 1)$, and $\mathbf{B}$ is the $L \times (L-1)$ matrix

$$\mathbf{B} = \begin{pmatrix} -1 & -1 & -1 & \cdots & -1 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

(a row of -1s atop the $(L-1) \times (L-1)$ identity).

We reframe the I-SSA problem under the cointegration constraint with simplified notation:

$$\mathbf{b} = \Gamma(0)\boldsymbol{\Xi}\mathbf{e}_1 + \boldsymbol{\Xi}\mathbf{B}\tilde{\mathbf{b}} = \mathbf{v} + \mathbf{P}\tilde{\mathbf{b}}, \tag{8}$$

with $\mathbf{b} = \mathbf{b}_\epsilon$, $\mathbf{v} = \Gamma(0)\boldsymbol{\Xi}\mathbf{e}_1$ and the full-column-rank matrix $\mathbf{P} = \boldsymbol{\Xi}\mathbf{B} \in \mathbb{R}^{L\times(L-1)}$. The MSE objective in (7) becomes:

$$J(\tilde{\mathbf{b}}) = \tilde{\mathbf{b}}'\tilde{\mathbf{A}}\tilde{\mathbf{b}} - 2\tilde{\mathbf{c}}'\tilde{\mathbf{b}} + \text{const}$$

where $\tilde{\mathbf{A}} = \mathbf{P}'\boldsymbol{\Sigma}'\boldsymbol{\Sigma}\mathbf{P}$, $\tilde{\mathbf{c}} = \mathbf{P}'\boldsymbol{\Sigma}'(\boldsymbol{\gamma} - \boldsymbol{\Sigma}\mathbf{v})$ where the constant depends on the MSE predictor $\boldsymbol{\gamma}_{\tilde{\Xi}\delta}$ which we rewrite as $\boldsymbol{\gamma} = \boldsymbol{\gamma}_{\tilde{\Xi}\delta}$.

The HT constraint can be rewritten as $\mathbf{b}'\mathbf{W}\mathbf{b} = 0$, where $\mathbf{W} = \mathbf{M} - \rho_1\mathbf{I}$. Substituting the cointegration constraint yields:

$$\tilde{\mathbf{b}}'\tilde{\mathbf{W}}\tilde{\mathbf{b}} + 2\tilde{\mathbf{u}}'\tilde{\mathbf{b}} + d = 0,$$

where $\tilde{\mathbf{W}} = \mathbf{P}'\mathbf{W}\mathbf{P}$, $\tilde{\mathbf{u}} = \mathbf{P}'\mathbf{W}\mathbf{v}$, and $d = \mathbf{v}'\mathbf{W}\mathbf{v}$. In the stationary SSA, feasibility assumes $|\rho_1| \leq \rho_{\max}(L)$, the largest eigenvalue of $\mathbf{M}$ or its spectral radius (Wildi (2026a), Proposition 1). As shown below, this feasibility condition extends to I-SSA. Because the boundary case $|\rho_1| = \rho_{\max}(L)$ renders the optimization 'trivial',[5] we restrict our focus to $|\rho_1| < \rho_{\max}(L)$. Under this assumption, $\mathbf{W}$ possesses both positive and negative eigenvalues, reducing the I-SSA problem to a GTRS with an indefinite equality constraint.

The Lagrangian is given by:

$$\mathcal{L}(\tilde{\mathbf{b}}, \lambda) = \tilde{\mathbf{b}}'\tilde{\mathbf{A}}\tilde{\mathbf{b}} - 2\tilde{\mathbf{c}}'\tilde{\mathbf{b}} + \lambda(\tilde{\mathbf{b}}'\tilde{\mathbf{W}}\tilde{\mathbf{b}} + 2\tilde{\mathbf{u}}'\tilde{\mathbf{b}} + d)$$

Setting the gradient with respect to $\tilde{\mathbf{b}}$ to zero yields the normal equations:

$$\tilde{\mathbf{H}}(\lambda)\tilde{\mathbf{b}}(\lambda) = \tilde{\mathbf{c}} - \lambda\tilde{\mathbf{u}}$$

where

$$\tilde{\mathbf{H}}(\lambda) = \tilde{\mathbf{A}} + \lambda\tilde{\mathbf{W}}$$

is the Hessian. Assuming $\tilde{\mathbf{H}}(\lambda)$ is invertible (invertibility is addressed below), the optimal $\tilde{\mathbf{b}}$ for a given λ is:

$$\tilde{\mathbf{b}}(\lambda) = \tilde{\mathbf{H}}(\lambda)^{-1}(\tilde{\mathbf{c}} - \lambda\tilde{\mathbf{u}}). \tag{9}$$

The I(1)-SSA solution to (7), under the cointegration constraint reparameterized via $\tilde{\mathbf{b}}(\lambda^*)$, follows from (9) with λ^* satisfying the secular equation

$$\mathbf{b}(\lambda^*)'\mathbf{W}\mathbf{b}(\lambda^*) = 0, \tag{10}$$

where $\mathbf{b}(\lambda^*) = \mathbf{v} + \mathbf{P}\tilde{\mathbf{b}}(\lambda^*)$. Below, we derive strict monotonicity—implying uniqueness—and a simple numerical solution to (10) via bisection.

Extensions to $d > 1$ follow by using $\mathbf{\Sigma}^d$ throughout, with additional cointegration constraints

$$\sum_{k=0}^{L-1} b_{\tilde{x}k}k^j = \sum_{k=0}^{L-1} \hat{\gamma}_{\tilde{x}\delta k}k^j, \quad j = 0, ..., d-1,$$

eliminating the higher-order unit root of $\tilde{x}_t$ in $e_t = y_{t,MSE} - y_t$ for I(d)-SSA; see Appendix (B). For $\lambda = 0$, I-SSA recovers the MSE predictor, noting that the cointegration constraint could be dropped since the MSE is already cointegrated.

3.3 Main Theoretical Results

The cointegration matrix $\mathbf{B}$ introduces asymmetries that render direct derivations feasible but needlessly complex. We therefore impose the cointegration constraint not a priori, but through an additional penalty term in the Lagrangian, maintaining $\Gamma(0) \neq 0$ and, for simplicity, setting $d = 1$ throughout (the proofs readily extend to $d > 1$). The problem formulation of the previous section nonetheless remains useful in deriving these results.

The first proposition delimits the region of strict positive definiteness of the Hessian $\mathbf{H}(\lambda)$ in the reformulated problem (Lagrangian), which does not depend on $\mathbf{B}$ anymore.

Proposition 1. *Assume $|\rho_1| < \rho_{\max}(L)$ (feasibility) in the HT constraint and define $\mathbf{A} = \mathbf{\Sigma\Xi}$ and $\mathbf{Q} = \mathbf{\Xi}'(\mathbf{M} - \rho_1\mathbf{I})\mathbf{\Xi}$. Consider the Hessian*

$$\mathbf{H}(\lambda) = \mathbf{A}'\mathbf{A} + \lambda\mathbf{Q},$$

and let $\nu_{\max}$ and $\nu_{\min}$ denote the maximum and minimum eigenvalues of the matrix $(\mathbf{A}'\mathbf{A})^{-1}\mathbf{Q}$. Then $\nu_{\max} > 0$ and $\nu_{\min} < 0$, and the matrix $\mathbf{H}(\lambda)$ is strictly positive definite if and only if λ lies in the open interval:

$$\lambda \in \left]-\frac{1}{\nu_{\max}}, -\frac{1}{\nu_{\min}}\right[=: J_{++}.$$

[5] In this case, the solution defaults to $\mathbf{b} = \pm c\mathbf{v}_{\rho\max}$, the corresponding eigenvector of $\mathbf{M}$, leaving only sign and scaling $c > 0$ for optimization (see Wildi, 2026a).

The proof is given in the Appendix. Strict convexity of $\mathbf{H}(\lambda)$ in J_{++}—termed 'hidden convexity'—allows us to formulate the main result characterizing the finite-length, $L < \infty$, I-SSA solution.

Theorem 1 (Algebraic Solution to I-SSA GTRS Problem)**.** *Consider the constrained optimization problem:*

$$\begin{aligned} \min_{\tilde{\mathbf{b}} \in \mathbb{R}^{L-1}} \quad & J(\tilde{\mathbf{b}}) = (\boldsymbol{\gamma} - \boldsymbol{\Sigma}\mathbf{b}_\epsilon)'(\boldsymbol{\gamma} - \boldsymbol{\Sigma}\mathbf{b}_\epsilon) \\ \textit{subject to} \quad & \mathbf{b}_\epsilon'\mathbf{M}\mathbf{b}_\epsilon = \rho_1 \mathbf{b}_\epsilon'\mathbf{b}_\epsilon \\ \textit{and} \quad & \mathbf{b}_\epsilon = \boldsymbol{\Xi}(\Gamma(0)\mathbf{e}_1 + \mathbf{B}\tilde{\mathbf{b}}). \end{aligned}$$

1. ***Stationary Point****: For $\lambda \in J_{++}$, a stationary point $\mathbf{b}_{\tilde{x}}(\lambda)$ of the Lagrangian is given by*

$$\mathbf{b}_{\tilde{x}}(\lambda) = \mathbf{H}(\lambda)^{-1}(\mathbf{A}'\boldsymbol{\gamma} - \mu(\lambda)\mathbf{1}),$$

with the scalar $\mu(\lambda)$ defined as

$$\mu(\lambda) = \frac{\mathbf{1}'\mathbf{H}(\lambda)^{-1}\mathbf{A}'\boldsymbol{\gamma} - \Gamma(0)}{\mathbf{1}'\mathbf{H}(\lambda)^{-1}\mathbf{1}},$$

where $\mathbf{1} \in \mathbb{R}^L$ is a vector of ones, and $\tilde{\mathbf{b}}(\lambda) = \mathbf{b}_{\tilde{x}}(\lambda)_{-0}$ (excluding the first lag-zero entry).

2. ***Feasibility****: For $|\rho_1| < \rho_{\max}(L)$ there exists $\lambda^* \in J_{++}$ such that*

$$\rho_{\mathbf{b}_\epsilon}(\lambda^*) = \frac{\mathbf{b}_\epsilon(\lambda^*)'\mathbf{M}\mathbf{b}_\epsilon(\lambda^*)}{\mathbf{b}_\epsilon(\lambda^*)'\mathbf{b}_\epsilon(\lambda^*)} = \rho_1,$$

with $\mathbf{b}_\epsilon = \boldsymbol{\Xi}\mathbf{b}_{\tilde{x}}$.

3. ***Uniqueness****: $\lambda^* \in J_{++}$ is uniquely determined by the HT constraint $\rho_{\mathbf{b}_\epsilon}(\lambda) = \rho_1$.*

4. ***Global Minimum****: the I-SSA solution $\mathbf{b}_{\tilde{x}}(\lambda^*)$, $\lambda^* \in J_{++}$, is the global minimum of the MSE objective under HT and cointegration constraints.*

5. ***Generalized Spectral Decomposition****: Let $\boldsymbol{\nu} = (\nu_1, ..., \nu_L)$ denote the generalized eigenvalues of the pencil $(\mathbf{A}'\mathbf{A}, \mathbf{Q})$ with corresponding generalized eigenvectors $\mathbf{v}_i$, $i = 1, ..., L$. Then*

$$\mathbf{b}_{\tilde{x}}(\lambda^*) = \sum_{i=1}^{L} \frac{w_i}{1 + \lambda^*\nu_i}\mathbf{v}_i, \tag{11}$$

where $\mathbf{w} = (w_1, ..., w_L)' = \mathbf{V}'(\mathbf{A}'\boldsymbol{\gamma} - \mu(\lambda^*)\mathbf{1})$ is the 'counterfactual' spectral factorization of the target.

6. ***AS-Dilemma****: The derivative of the MSE objective function $J(\lambda)$ with respect to λ, $\lambda \in J_{++}$, is*

$$\frac{dJ(\lambda)}{d\lambda} = -\lambda\frac{dF(\lambda)}{d\lambda}, \tag{12}$$

where $F(\lambda) = \mathbf{x}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda)$ is the secular function with $\frac{dF(\lambda)}{d\lambda} < 0$ (strict monotonicity). This can be decomposed into an autocorrelation-driven term and a scale-driven term:

$$\frac{dJ(\lambda)}{d\lambda} = \underbrace{-\lambda\|\mathbf{b}_\epsilon(\lambda)\|^2\frac{d\rho_{\mathbf{b}_\epsilon}(\lambda)}{d\lambda}}_{\textit{Autocorrelation effect}} \underbrace{-\lambda\big(\rho_{\mathbf{b}_\epsilon}(\lambda) - \rho_1\big)\frac{d\|\mathbf{b}_\epsilon(\lambda)\|^2}{d\lambda}}_{\textit{Scale effect}}. \tag{13}$$

At the root $\lambda = \lambda^*(\rho_1)$ of the secular equation $F(\lambda^*) = 0$ we have

$$\frac{dJ(\lambda)}{d\lambda}\bigg|_{\lambda^*(\rho_1)} = \underbrace{-\lambda^*(\rho_1)\|\mathbf{b}_\epsilon(\lambda^*(\rho_1))\|^2\frac{d\rho_{\mathbf{b}_\epsilon}(\lambda)}{d\lambda}}_{\textit{Autocorrelation effect}}\Bigg|_{\lambda^*(\rho_1)}, \tag{14}$$

and

$$\left.\frac{dJ(\lambda)}{d\rho_1}\right|_{\lambda=\lambda^*(\rho_1)} = -\lambda^*(\rho_1)(\mathbf{b}^*_\epsilon)'\mathbf{b}^*_\epsilon = \begin{cases} <0 & \rho_1 < \rho_{MSE} \\ >0 & \rho_1 > \rho_{MSE} \end{cases}, \tag{15}$$

with $\mathbf{b}^*_\epsilon := \mathbf{b}_\epsilon(\lambda^*(\rho_1))$.

A proof is given in the Appendix.

The strict monotonicity of $F(\lambda)$ (Assertion 6) ensures the root $F(\lambda^*) = 0$ is obtainable via efficient numerical optimization, e.g., bisection of J_{++}. Note that the theorem is essentially assumption-free since its proof relies only on generally valid algebraic identities, assuming $\mathbf{\Xi}$ to be invertible, which is ensured by $\xi_0 = 1$ on its main diagonal (even the standing assumption $\Gamma(0) \neq 0$ could be dropped). However, an important caveat concerns the interpretation of the HT constraint and the objective function. Both rely on finite MA approximations, justifying, e.g., (6), which links the objective function (right-hand side) to the effective MSE (left-hand side). For these approximations to be valid, the predictor weights must decay sufficiently fast. This gap is closed by Theorem 2, which provides the missing restrictions and assumptions. For now, we assume that the objective and the constraint admit their intended interpretations. A more thorough discussion is provided in Appendix A, contrasting I-SSA with SSA.

The following corollary 'explains' the typical tipped or nose-shaped profile of the I-SSA predictor at the left boundary (early lags), as illustrated in Section 4, as well as the decay of filter coefficients with increasing lag in typical applications[6], signifying stability of the difference equation despite the presence of non-stationary operators.

Corollary 1. *Assume $\rho_1 \neq \rho_{MSE}$ (non-degenerate case). Then the solution of Theorem 1 can be written as a 'time domain' discrete difference equation*

$$\big[\mathbf{\Sigma}'\mathbf{\Sigma} + \lambda(\mathbf{M} - \rho_1\mathbf{I})\big]\mathbf{v} = \mathbf{d}, \tag{16}$$

where $\mathbf{v} = \mathbf{\Xi}\mathbf{b}_{\tilde{x}}$ and $\mathbf{d} = \mathbf{\Sigma}'\boldsymbol{\gamma} - \mu(\lambda)(\mathbf{\Xi}')^{-1}\mathbf{1}$. This difference equation has implicit Dirichlet and 'extended' Dirichlet boundary constraints at the left and right boundaries. Specifically, the 'ghost' values $b_{\tilde{x},-1}, b_{\tilde{x},L}$ implied by these constraints satisfy

$$b_{\tilde{x},-1} = 0 \quad \textit{and} \quad b_{\tilde{x},L} = -\sum_{j=0}^{L-1} \xi_{j+1} b_{\tilde{x},L-(j+1)},$$

where ξ_k are the AR-inversion weights defined by $\mathbf{\Xi}^{-1}$.

The proof is in the Appendix A with a background discussion. Note that $\mathbf{\Xi}^{-1}$ always exists since $\xi_0 = 1$ on the main diagonal.

The next theorem establishes asymptotic convergence of the I-SSA solution as $L \to \infty$ and shows that the I-SSA coefficients decay fast enough to justify finite MA approximations under fairly general assumptions on target and Wold weights. For this purpose, let ℓ^2 denote the Hilbert space of square-summable real sequences $\mathbf{x} = (x_0, x_1, \dots)'$ with $\|\mathbf{x}\|_2^2 = \sum_{k\geq 0} x_k^2 < \infty$. Finite-length vectors $\mathbf{x} \in \mathbb{R}^L$ are identified with elements of ℓ^2 by zero-padding, and an infinite matrix (operator) is called bounded on ℓ^2 if $\|\mathbf{A}\| := \sup_{\mathbf{x}\neq 0} \|\mathbf{A}\mathbf{x}\|_2/\|\mathbf{x}\|_2 < \infty$.

Theorem 2 (Asymptotic I-SSA and Interpretability)**.** *Under the standing assumptions $\Gamma(0) \neq 0$ (preservation of integration order, together with invertibility of the Wold decomposition, see below) and $\rho_1 \neq \rho_{MSE}$ (non-degeneracy), and for a given target $\hat{\boldsymbol{\gamma}}_{MSE}$ and Wold decomposition $\boldsymbol{\xi}$, assume there exists $L_0 > 0$ such that $|\rho_1| < \rho_{\max}(L_0)$, the largest eigenvalue of $\mathbf{M}(L_0)$. For $L > L_0$, consider the I-SSA predictor $\mathbf{b}^*_{\tilde{x}} := \mathbf{b}_{\tilde{x}}(\lambda^*)$ from Theorem 1, with weights b^*_k, $k = 0, \dots, L-1$. Suppose there exist $s > 1$ and constants $C_\xi, C_\gamma > 0$, independent of L, such that:*

[6] Excluding, e.g., excessive smoothing (or un-smoothing) by an extremal HT constraint $|\rho_1| \to \rho_{\max}(L)$.

(A1) $|\xi_j| \le C_\xi j^{-s}$ *for* $j = 0, \dots, L-1$ *(polynomial decay of MA weights), and* $\xi(z) \neq 0$ *for* $|z| \le 1$ *(invertibility);*

(A2) $|\gamma_{MSE,k}| \le C_\gamma k^{-s}$ *for* $k = 0, \dots, L-1$ *(polynomial decay of the target);*

(A3) The sequence $\lambda^*(L)$ *of optimal Lagrange parameters in Theorem 1 is uniformly bounded away from the boundary of* $J_{++}(L)$*, i.e.,* $\inf_L$ *dist*$(\lambda^*(L), \partial J_{++}(L)) > 0$.

Then, as $L \to \infty$*:*

(i) The boundaries of the strict positivity domain, $\lambda_{\min}(L)$ *and* $\lambda_{\max}(L)$*, converge to finite limits*

$$\begin{aligned} \lambda_{\min}(L) &\to -\frac{2}{(1-\rho_1)^2} < 0 \\ \lambda_{\max}(L) &\to \frac{1}{4(1+\rho_1)} > 0. \end{aligned}$$

(ii) There is a constant C^**, depending only on* $(s, C_\xi, C_\gamma, \rho_1)$ *and the uniform separation bound, but not on* L*, such that for all* L *and all* $k = 0, \dots, L-1$*:*

$$|b_k^* - \gamma_{MSE,k}| \le C^* k^{-s} \qquad \textit{and hence} \qquad |b_k^*| \le (C_\gamma + C^*) k^{-s}.$$

In particular, the weights $\mathbf{b}_x^* = (b_0^*, \dots, b_{L-1}^*)'$ *decay polynomially at rate* s*, uniformly in* L.

(iii) The sequence of optimal Lagrange multipliers $\lambda^*(L)$ *converges to a unique limit* λ_∞^* *strictly inside* $J_{++}(\infty)$*, satisfying the asymptotic secular equation*

$$\mathbf{b}_\infty(\lambda_\infty^*)' \mathbf{Q}_\infty \mathbf{b}_\infty(\lambda_\infty^*) = 0, \quad \textit{where} \quad \mathbf{Q}_\infty = \boldsymbol{\Xi}_\infty'(\mathbf{M}_\infty - \rho_1 \mathbf{I}_\infty)\boldsymbol{\Xi}_\infty,$$

and

$$\begin{aligned} \mathbf{b}_\infty(\lambda_\infty^*) &= \mathbf{d}_\infty(\lambda_\infty^*) + \hat{\boldsymbol{\gamma}}_{MSE\infty} && (17) \\ \mathbf{d}_\infty(\lambda_\infty^*) &= -\lambda_\infty^* \boldsymbol{\Xi}_\infty^{-1} \boldsymbol{\Delta}_\infty \mathbf{K}_\infty(\lambda_\infty^*)^{-1} \mathbf{f}_\infty \\ \mathbf{f}_\infty &= \boldsymbol{\Delta}_\infty'(\mathbf{M}_\infty - \rho_1 \mathbf{I}_\infty)\boldsymbol{\Xi}_\infty \hat{\boldsymbol{\gamma}}_{MSE\infty} \\ \mathbf{K}_\infty(\lambda_\infty^*) &= \mathbf{I}_\infty + \lambda_\infty^* \boldsymbol{\Delta}_\infty'(\mathbf{M}_\infty - \rho_1 \mathbf{I}_\infty)\boldsymbol{\Delta}_\infty, && (18) \end{aligned}$$

where $\boldsymbol{\Xi}_\infty$*,* $\boldsymbol{\Delta}_\infty$*,* $\mathbf{I}_\infty$*,* $\mathbf{M}_\infty$ *are the one-sided infinite versions of* $\boldsymbol{\Xi}_L$*,* $\boldsymbol{\Delta}_L$*,* $\mathbf{I}_L$*,* $\mathbf{M}_L$*, and all operators are bounded on* ℓ^2*; in particular, the modified Hessian* $K_\infty(\lambda_\infty^*)$ *is also invertible. Finite-length solutions, embedded in* ℓ^2 *by zero-padding, converge in* ℓ^2*-norm to the infinite-length solution.*

The proof is in the appendix, together with a background discussion that covers, among other things, the modified and bounded Hessian $\mathbf{K}_L(\lambda)$, the assumptions, and their verification in applications. The last Corollary specifies the efficient AS frontier of I-SSA.

Corollary 2. *Let the assumptions of Theorem 2 hold. Then, for* $\rho_1 > \rho_{MSE}$*, the optimized MSE objective of the I-SSA is strictly increasing in* ρ_1*, and no linear predictor of the same (or smaller) length* L *can both reduce MSE (in levels) and increase* ρ_1 *in differences: for Gaussian processes with vanishing drift, I-SSA is maximally monotone, in the sense that for given MSE tracking accuracy (on non-stationary levels), no other linear predictor of that (or smaller) length achieves a smaller TP rate—defining the 'right branch' of the efficient frontier (*$\rho_1 > \rho_{MSE}$*). The reverse holds for* $\rho_1 < \rho_{MSE}$*: no linear predictor of the same (or smaller) length can reduce both MSE and* ρ_1*, and I-SSA is minimally monotone, in the sense that for given MSE tracking accuracy (on non-stationary levels), no other linear predictor of that (or smaller) length achieves a larger TP rate—defining the 'left branch' of the efficient frontier (*$\rho_1 < \rho_{MSE}$*).*

The proof is in the Appendix. Conditioning on Gaussianity is generally not a limitation, since Eq. (3) linking HT and first-order ACF is fairly robust to departures from Gaussianity—particularly under accentuated smoothness, where smoothing tips 'naturally' toward Gaussianity via the central limit theorem (Wildi 2026b). In case of a non-vanishing drift, the Corollary addresses the crossover rate about the linear trend. In applications, however, the drift, i.e., the mean of stationary first differences, is often negligible relative to the stationary growth-dynamics above and below the drift, so that the (mean) distance between consecutive zero-crossings (TPs) is identical or similar to that between consecutive mean-crossings (crossover points), effectively addressed by the HT. We do not treat the case of asymmetry, with crossings of a line markedly above or below the mean, since this would break the symmetries induced by the first-order ACF. In any case, I-SSA maximizes smoothness in terms of the first-order ACF of stationary differences, retaining interpretability and affecting the crossing rate effectively at any level.

4 Applications: Real-Time Business-Cycle Analysis and I-SSA Trend

We first apply I-SSA to HP, customizing the filter on the novel AS frontier. The data is the US INDPRO (industrial production) series, and the target is the two-sided HP filter; we compute an optimal MSE predictor on levels replicating the TP rate of the one-sided HP in differences. We expect the I-SSA nowcast to outperform the classic concurrent HP in tracking the two-sided HP while matching the latter's smoothness. In the second example, the target is the original data instead of the two-sided HP, illustrating the novel I-SSA trend (and smoothing) concept. We apply the design to a random-walk process and to INDPRO, benchmarking against HP and the boosted HP (Mei, Phillips and Shi, 2024), imposing the respective TP rates on I-SSA. For identical smoothness, I-SSA outperforms in terms of MSE, specifically at peaks and dips of the series, where the nowcast is faster (advanced) without over- or undershooting, which makes it well suited to real-time assessment.

4.1 HP Customization: Maximal Monotone HP Trend-Nowcast

We apply the I(1)-SSA predictor $\mathbf{b}_{\tilde{x}} = \Gamma(0)\mathbf{e}_1 + \mathbf{B}\tilde{\mathbf{b}}$, where $\tilde{\mathbf{b}}$ is determined by Equation (9), to the monthly US-industrial production index INDPRO[7], as illustrated in Fig.(1). The target is specified using a two-sided HP filter with parameter λ =14400 for monthly data, assuming a nowcast (i.e., $\delta = 0$) and $L = 201$. We benchmark the I(1)-SSA approach against the MSE design, positing that the differenced data adheres to an AR(1)-model (results are robust against alternative MA or ARMA specifications). This assumption is supported by the weak, somewhat unsystematic, yet slightly persistent ACF pattern observed (see the bottom right panel of the figure). The AR(1) model is utilized to derive the weights $\boldsymbol{\xi}$ in the Wold decomposition of the (differenced) time series. For a comprehensive analysis, we also incorporate the classic one-sided HP concurrent filter, referred to as HP-C, as an additional benchmark, see McElroy (2008) and Cornea-Madeira (2017) for background.

[7]Board of Governors of the Federal Reserve System (US), Industrial Production: Total Index [INDPRO], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/INDPRO, October 31, 2024.

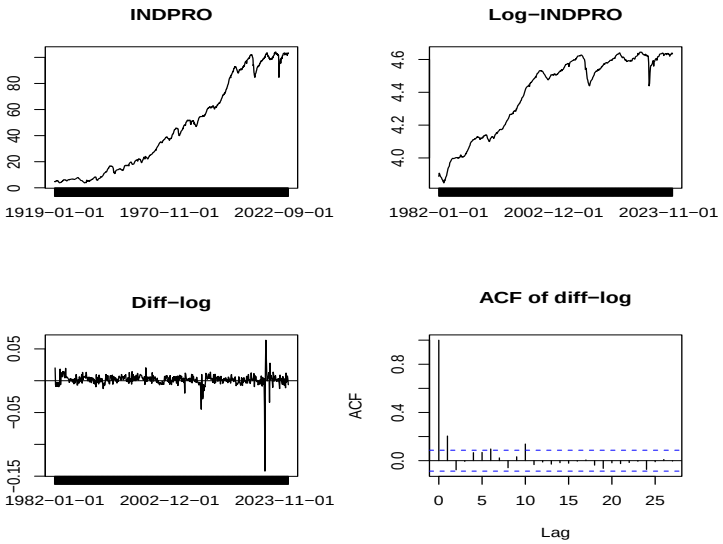


Figure 1: INDPRO original entire sample (top left), log-transformed INDPRO from 1982 onwards (top right), log-differenced data (bottom left) and ACF of log-differenced series (bottom right).

All (trend-) filters are displayed in Fig. (2). The two-sided (truncated) HP filter, with coefficients γ_k for $|k| \leq (L-1)/2$, is symmetrically centered at lag $k = (L-1)/2$ (violet line, left panel). The nowcast presumes that the differenced data follow the AR(1) specification above, with performance improving relative to the WN hypothesis. The classic MSE predictor (green) allocates most weight to the most recent observations, owing to the smooth trajectory of the associated ARIMA(1,1,0) process, which requires little additional noise suppression by the nowcast filter. This illustrates that a unilateral focus on MSE may yield 'noisy' predictors, as shown in Fig. (4).

The first-order ACF of the differenced MSE nowcast is $\rho_{MSE} = 0.105$, corresponding to $HT = 2.143$, barely above white noise. In contrast, the first-order ACF of the HP-C filter, $\rho_{HP-C} = 0.954$, is substantially higher, with $HT_{HP-C} = 10.264$, and we therefore set $ht_1 = HT_{HP-C}$ in the HT constraint. The resulting optimal Lagrange multiplier is $\lambda^* = -102.573$; its negative sign and magnitude indicate a markedly stronger smoothing effect than the MSE benchmark, as desired. The effective first-order ACF $\rho^* = 0.954$ of $\mathbf{b}_\epsilon(\lambda^*) = \boldsymbol{\Xi}\left(\Gamma(0)\mathbf{e}_1 + \mathbf{B}\tilde{\mathbf{b}}(\lambda^*)\right)$ matches the HT constraint, confirming convergence of the numerical optimization. In summary, I(1)-SSA replicates the smoothness (TP rate) of HP-C while improving tracking through a lower MSE on the AS-frontier.

The bottom panel of Fig. 2 shows the weights of $\mathbf{H}(\lambda^*)^{-1}$ assigned to the MSE target for $\lambda^* = -102.573$ and at lags $k = 0, 10, 20, \ldots, L-1 = 100$ (multiples in ten) noting that $b^*_{\tilde{x},k-1} = \mathbf{H}(\lambda^*)^{-1}_{k\cdot}\left(\mathbf{A}'\boldsymbol{\gamma} - \mu(\lambda^*)\mathbf{1}\right)$. The figure shows that the k-th row $\mathbf{H}(\lambda^*)^{-1}_{k\cdot}$ of the inverted Hessian is concentrated at lag k, with symmetric decay in the middle ($k = 50$) and asymmetric decay towards the boundaries $k = 0$ and $k = L-1$.

Assumption (A3) of Theorem 2 concerns the sequence $\{\lambda^*(L)\}_L$. In our application it holds numerically: $\lambda^*(L) \simeq -102.573$ is essentially constant in L for $L > 100$ and well away from the finite-sample boundaries $(-944.87, 0.13)$ of J_{++} as well as from the asymptotic boundaries $(-925.92, 0.128)$ given by the first assertion of Theorem 2. The proximity of finite and asymptotic boundaries confirms near convergence. Furthermore, the extreme eigenvalues of the 'stationary' Hessian $\mathbf{K}(L)$ stabilize rapidly at $\lambda_{\min} = 1.685 > 0$ and $\lambda_{\max} = 474.7$, confirming invertibility and boundedness, respectively (see the accompanying I-SSA package). This is consistent with the fast decay of the I-SSA predictor weights —implied by the theorem— in the top two panels of the

figure, and supports the validity of the finite MA approximation underlying (6).

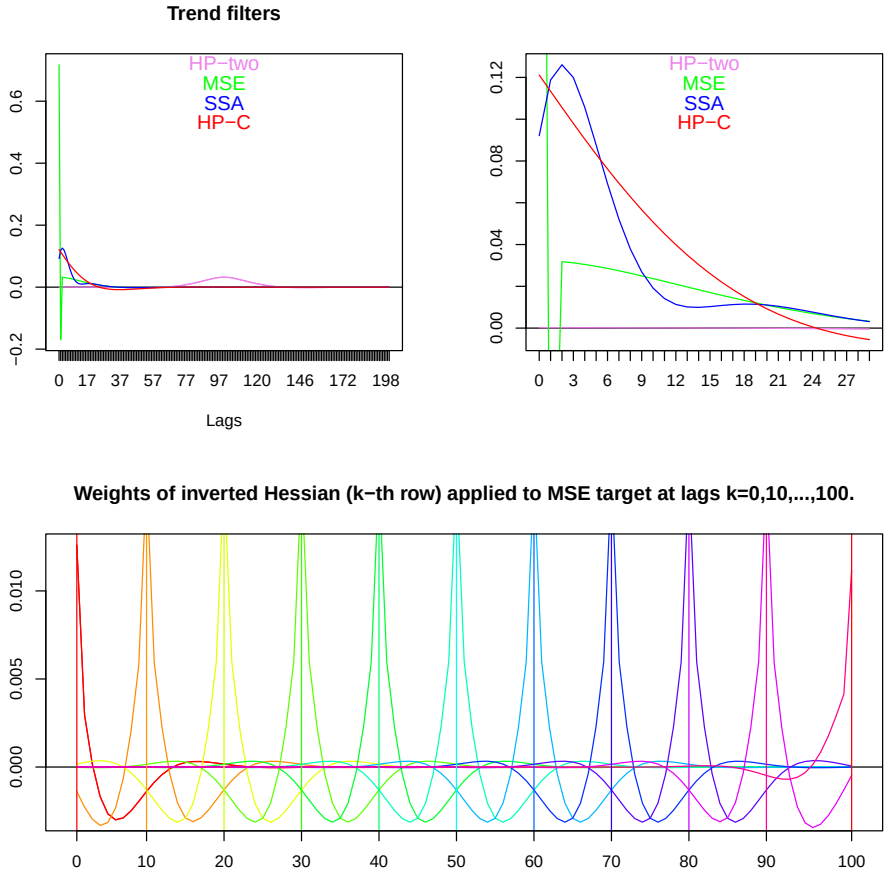


Figure 2: Top panels: two-sided HP target (violet), HP-C (red), MSE (green), and I(1)-SSA (blue); the latter two are based on an ARIMA(1,1,0) specification. All filter coefficients sum to one (cointegration constraint). The left panel shows all lags; the right panel shows the first thirty, with the vertical axis truncated. I-SSA exhibits the typical nose shape, peaking at lag $k = 2$ in this example, consistent with Dirichlet boundary constraints. Bottom panel: weights assigned by $\mathbf{H}(\lambda^*)^{-1}$, with $\lambda^* = -102.57$, to the MSE target at lags $k = 0, 10, \ldots, L-1$. The $k+1$-th row of the inverted Hessian is concentrated at lag k, with symmetric decay in the middle ($k = 50$) and asymmetric decay towards the boundaries $k = 0, k = L - 1 = 100$.

The non-stationary (logarithmic) INDPRO series (black), along with the filter outputs, is illustrated in Fig.(3), covering a historical timeframe spanning the last three recessions, during which the index lost previous momentum. The acausal two-sided HP filter (violet) cannot extend to the sample's actual endpoint. I(1)-SSA (blue) is characterized as maximal monotone for given MSE tracking, or as maximal tracking for a given TP rate. The figure shows that HP-C (red) tracks the symmetric HP better in the middle of longer up- or downturns but is prone to over- and undershooting at peaks and troughs, where it also lags I-SSA, yielding a net MSE loss quantified in Table (1). If the series were effectively I(2), with up- or down-swings of unbounded length asymptotically, HP-C would outperform I-SSA in MSE; but increasing $d = 1$ to $d = 2$ would then re-establish I-SSA's outperformance, sitting on the adjusted frontier.

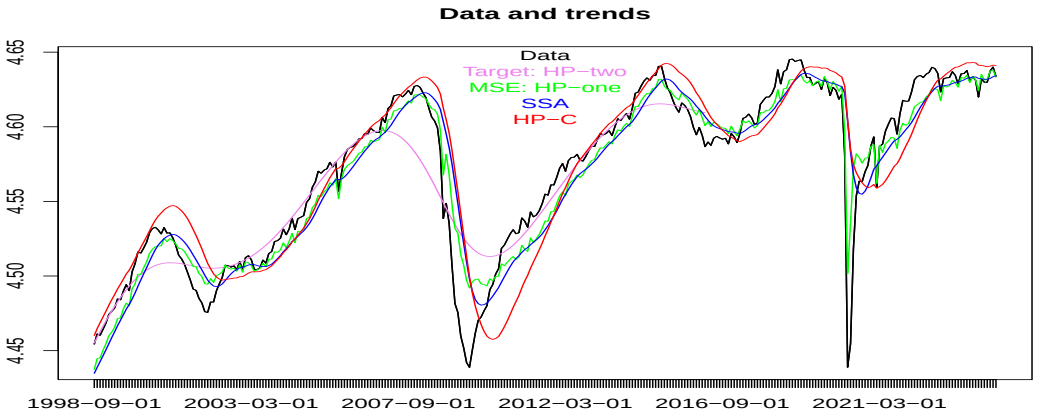


Figure 3: (Log-) INDPRO (black), two-sided HP target (violet), MSE (green), HP-C (red), and I(1)-SSA (blue). HP-C tracks the symmetric HP better in the middle of longer up- or downturns but is prone to over- and undershooting relative to the HP target at peaks and troughs, where it also lags I-SSA, yielding a net MSE loss. .

Table (1) summarizes sample performances in MSE, referenced against the two-sided HP filter, and empirical HTs of the differenced predictors. HP-C exhibits the lowest accuracy, with empirical MSE exceeding twice the optimal benchmark. I(1)-SSA, in between, confirms the anticipated MSE hierarchy. Exchanging MSE for TP rate along the efficient AS frontier incurs minimal tracking loss, with I(1)-SSA's TP rate seemingly surpassing HP-C's[8]. This trade-off is justified by the analysis's specific objectives: given its prevalence in applications, the smoothness offered by HP-C appears more desirable than pure MSE performance—otherwise a rational analyst would resort to the latter. We contend that I(1)-SSA can enhance accuracy while making up in smoothness. For this target and series, a stringent focus on MSE-accuracy, as in classic signal extraction, cannot be achieved without substantial TP-rate smoothness losses, as evidenced by the MSE predictor's performance.

	MSE	SSA	HP-C
Sample mean square error	0.00024	0.00037	0.00050
Sample holding time	2.30597	14.04545	12.87500

Table 1: Sample performances of MSE, HP-C and I(1)-SSA predictors: mean-square error relative to the acausal two-sided HP (first row) and empirical holding times of the stationary differenced predictors (second row).

[8]The discrepancy in sample HTs between I(1)-SSA and HP-C, despite identical first-order ACFs, may stem from the limited number of zero-crossings, affecting the reliability of sample estimates. Both empirical holding times can be matched with arbitrary fidelity for sufficiently long simulated samples within our I-SSA package.

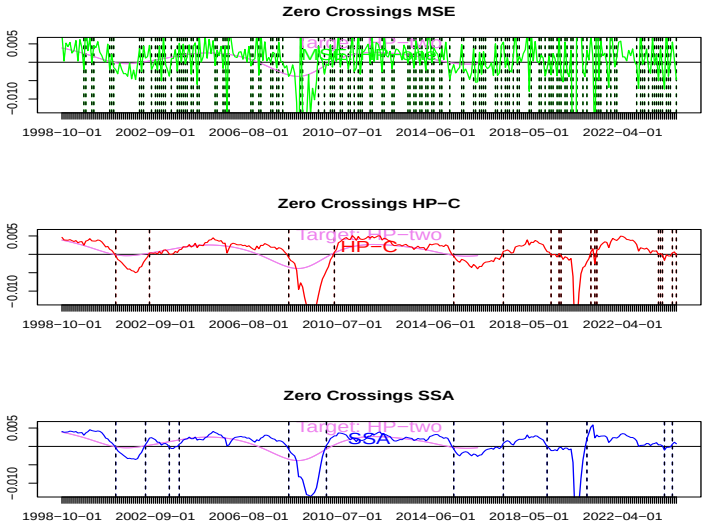


Figure 4: Differenced filter outputs: two-sided target (violet), MSE (green), HP-C (red) and I(1)-SSA (blue) with zero-crossings of the designs marked by vertical lines. MSE (top), HP-C (middle) and I(1)-SSA (bottom).

To underscore this point, Fig.(4) visualizes the HT discrepancy by juxtaposing the outputs of the differenced filters with the zero-crossings of each design, marked by corresponding vertical lines.

Our findings suggest that smoothness can be substantially improved over the MSE benchmark without overtly sacrificing accuracy. More precisely, the marginal cost of increased TP rate, in terms of increased MSE, is characterized by the AS-dilemma (15) (using 3 to link ACF and HT). Fig.(4) further shows that the last three recession episodes are effectively tracked through the first differences of the smooth HP-C and I-SSA nowcasts, with minimal 'false alarms' during extended expansions, noting that TP dating by I-SSA is either coincident or advanced. In short, the proposed I(1)-SSA framework reconciles first-difference analysis (zero-crossings associated with TPs) and optimal level tracking within a single, logically consistent, statistically efficient, and interpretable design. It addresses MSE and HT simultaneously, as well as target correlation and sign accuracy through the stationary proxy predictor on the right of (6).

4.2 I-SSA Trend

In contrast to the previous section, the target here is the original series (identity filter $\mathbf{I}$ instead of HP): I-SSA tracks the level of the data as closely as possible subject to an HT constraint on first differences. This resembles classical Whittaker-Henderson smoothing, though the penalty term is the TP rate rather than curvature or mean-squared second-order differences.

For illustration, we rely on a random-walk process—$d = 1$ and $\boldsymbol{\Xi} = \mathbf{I}$—covering both the mid-sample acausal symmetric trend smoother and the real-time causal one-sided trend nowcast (similar findings apply to $d > 1$ or $\boldsymbol{\Xi} \neq \mathbf{I}$, not shown). The design is then applied 'as is' to the INDPRO series used previously. I-SSA can accommodate a formulated model of the data through d and $\boldsymbol{\Xi}$, but the random-walk setting conforms well with typical economic prediction problems (see the accompanying R-package for background), justifying a 'model-free' I-SSA trend based on these default settings.

1) Random-Walk:

We compare I-SSA to HP and I-SSA-B to the boosted variant, abbreviated, HP-B, as implemented in the R-package *MacroFilters*. For the random walk, the HP-B algorithm selected 100 iterations for $\lambda = 129600$ (monthly setting), based on BIC. Empirical performances are based on a single very long sample, of length 10^7, and we compare the I-SSA two-sided smoothers and one-sided nowcasts, each replicating the respective true HTs of HP and HP-B on white noise (differenced random-walk), namely 59.51 (HP-two), 7.651 (HP-one), 30.28 (HP-B-two) and 5.633 (HP-B-one).

Performance is reported in Tables 2 (two-sided acausal smoothers) and 3 (one-sided causal nowcasts). We verify throughout that the sample HT converge to the expected HT. HP-B designs are more adaptive (smaller HT and MSE, larger curvature) than the original HP, as expected. I-SSA-two reduces MSE by roughly 50% relative to HP-two for identical HT and nearly identical curvature; I-SSA-B-two reduces MSE by roughly 7% for identical HT and virtually identical curvature—in both cases a worthwhile trade-off. I-SSA-two has slightly smaller ZC-distance variance than HP (marginally more regular), while I-SSA-B-two is marginally more irregular. HP's twice-as-large HT corresponds to roughly a four-times-larger variance than HP-B's (similarly for I-SSA).

Table 3 shows that two-sided designs exert much stronger smoothing, with HP's HT nearly 8 times larger than its nowcast's, yielding an MSE that even exceeds the nowcast's—hurting a priori expectations, since the two-sided design 'knows' the future. In contrast, HP-B-two has smaller MSE than its nowcast, in agreement with expectations. However, I-SSA-B-one nowcast has substantially smaller MSE still. In general, I-SSA nowcasts reduce MSE by roughly 50% for identical HT but at a cost of roughly 25% larger curvature—a worthwhile trade-off, particularly when tracking matters. Marginal costs, derived from Eq.(15), quantify a 8.796% increase in MSE for a 1% increase in ρ_1, equivalently a 1.569% MSE increase per 1% increase in HT, for I-SSA-one; for I-SSA-B-one the corresponding losses are 5.46% and 1.899%, allowing for more informed trade-offs on the frontier than just replicating the benchmarks' HTs (not pursued here). Interestingly, I-SSA-one outperforms HP-B-one on all three measures.

In general, the ZC-distance variance of I-SSA nowcasts is substantially smaller than that of HP and HP-B, indicating more evenly spaced TPs. This can be explained by the filter profiles in Fig. 5 (we highlight HP-B; HP is similar): faster decay (greater adaptivity, less phase-lag and retardation, i.e., systematic left-shift) and a stronger side lobe (needed to preserve equal HT) let one-sided I-SSA track peaks and dips of the series faster and more tightly, at an identical TP rate (bottom panel), with less hesitation, thereby avoiding spurious ('noisy') TPs around them; all are useful features in a real-time dating exercise. In contrast, the curvature concept underlying HP, and thus HP-B, imposes less inflection, possibly conflicting with the irregular dynamics of random-walk-like series, i.e., series lacking overt smoothness. This places 'true' TPs farther apart while clustering spurious ones around them (larger variance, more irregular dating), especially since HP's implicit ARIMA(0,2,2) representation (McElroy, 2008) has a TP rate that vanishes asymptotically. Given this structural unit-root misspecification implied by HP's minimal-curvature concept, together with the measured MSE loss, we may prefer the more regular and evenly spaced I-SSA TP dating for series lacking the strong intrinsic smoothness entailed by I(2) processes. For the latter, we recommend increasing the integration order to $d = 2$ in I-SSA.

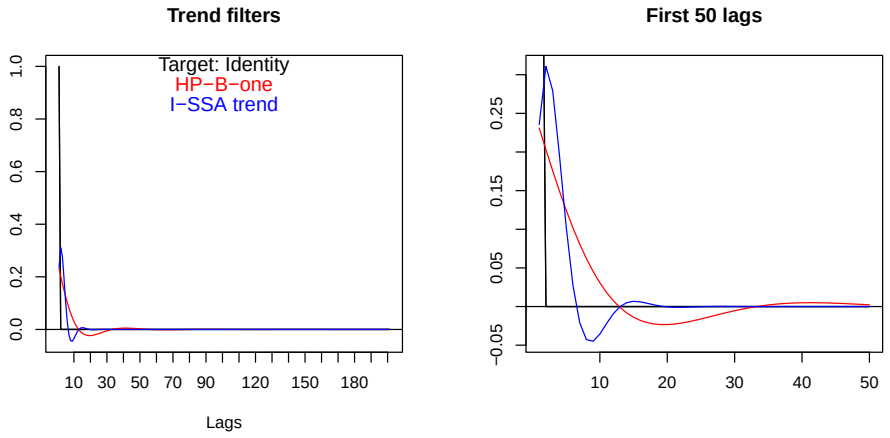
Evidently, our comparisons are biased against HP,HP-B, since I-SSA relies on knowledge of the true model. The results above thus indicate what is left on the table by ignoring the true DGP, assuming an identical TP rate across indicators.

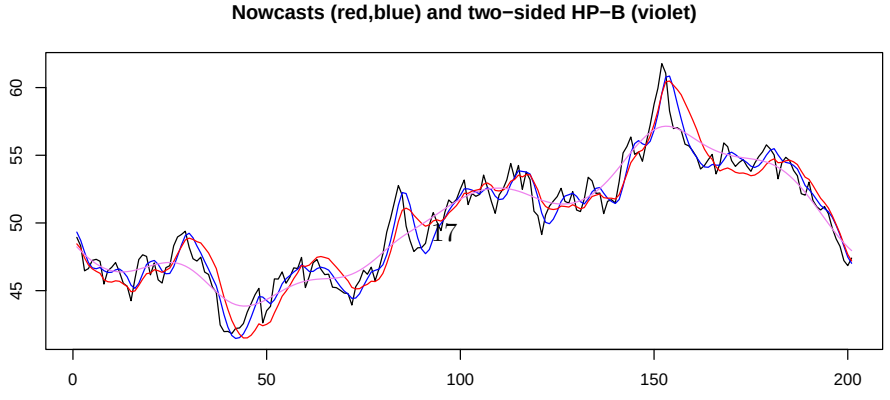
	HP-two	I-SSA-two	HP-B-two	I-SSA-B-two
MSE	5.94693	2.83754	1.52707	1.43873
Empirical HT	59.51446	59.51729	30.30022	30.28829
Curvature	0.00820	0.00868	0.02376	0.02392
Variance of ZC distance	1870.07671	1708.17217	435.67544	439.23862

Table 2: Acausal two-sided smoothers: sample performance of HP-two, I-SSA-two (replicating HP's HT), and HP-B-two, I-SSA-B-two (replicating HP-B's HT) for a single, very long realization of the random walk. The sample HT replicates the expected HT up to negligible sample variation. HP-B is more adaptive than HP, with smaller MSE and HT but larger curvature. I-SSA-two reduces MSE by roughly 50% for identical HT and nearly identical curvature; I-SSA-B-two reduces MSE by roughly 7% for identical HT and virtually identical curvature. I-SSA-two has slightly smaller ZC-distance variance than HP (marginally more regular); I-SSA-B-two is marginally more irregular. HP's twice-as-large HT corresponds to roughly a four-times-larger variance.

	HP-two	HP-one	I-SSA-one	HP-B-two	HP-B-one	I-SSA-B-one
MSE	5.94693	3.39404	1.50722	1.44740	1.62800	0.88949
Empirical HT	59.51482	7.65129	7.65620	30.30282	5.62714	5.63246
Curvature	0.00820	0.12404	0.18577	0.02376	0.24184	0.29457
Variance	1870.12055	96.15100	31.87054	435.72189	34.90346	16.98246

Table 3: Causal one-sided smoothers (two-sided HP, HP-B included for reference, first and fourth columns): sample performance of HP-one, I-SSA-one (replicating HP's HT), and HP-B-one, I-SSA-B-one (replicating HP-B's HT) for a single, very long realization of the random walk. The sample HT replicates the expected HT up to negligible sample variation. Two-sided designs exert much stronger smoothing, with HP's HT nearly 8 times larger than its nowcast's, resulting in an MSE exceeding the nowcast's. In contrast, HP-B-two has smaller MSE than its nowcast, while I-SSA-B-one has substantially smaller MSE still. In general, I-SSA reduces MSE by roughly 50% for identical HTs but larger curvature. Interestingly, I-SSA-one outperforms HP-B-one on all three measures. The ZC-distance variance of the I-SSA nowcasts is substantially smaller than that of the HP and HP-B nowcasts, indicating markedly greater regularity of real-time TP dating (discussed in the main text).





2) INDPRO:

We apply the above filters 'as is,' in their default settings, to INDPRO, assessing the effects of potential model misspecification on I-SSA. We focus on HP-B, a more challenging benchmark for I-SSA, and on nowcasts (additional comparisons are offered in the accompanying R-package). When applied to white noise, the HT of the one-sided HP-B is $HT \approx 5.63$ (see Table 3). However, differences of log-INDPRO are weakly autocorrelated (Fig. 1), so we also provide a corrected, slightly larger value, $HT \approx 7.43$, obtained by applying HP-B to an AR(1) with $a_1 = 0.3$ from the previous section. This correction affects the optimization only through ht_1 in the constraint; we still assume $\mathbf{\Xi} = \mathbf{I}$ (white noise).

Table 4 reports empirical MSE, HT, and Curvature statistics for the data in Section 4.1. The HT of HP-B lies between the two I-SSA values, confirming weak positive dependence in the series, though the AR(1) correction seems slightly conservative (excessive smoothness). In any case, both I-SSA filters outperform HP-B, reducing MSE by roughly 25% and 50%, respectively. Interestingly, the smoother I-SSA also outperforms in terms of Curvature. Table 5 reports the true (expected) MSE (based on optimized objective functions) and HT for I-SSA, under the assumption of the RW model. The slight positive bias in the empirical numbers confirms the misspecification: if the differenced data were truly white noise, the smoothed series would exhibit slightly more zero crossings and a slightly smaller MSE.

Fig. 6 compares filter coefficients (top) and smoothed series (bottom); (blue and red filter coefficients are the same as in Fig.5). Increasing the HT in I-SSA (green vs. blue) shifts the weights slightly to the right, with slower decay and a right-shifted side lobe positioned between the faster I-SSA and HP-B. The smoothed series in the bottom panel suggest a tighter tracking of the data at peaks and dips, where both I-SSA generally lead HP-B, without over- or under-shooting.

This example illustrates that I-SSA based on the default random-walk specification can perform well on economic time series, provided first differences are not too far away from white noise. In the presence of weak autocorrelation, we recommend a slight adjustment of the HT constraint to correct for the resulting bias. In the case of stronger dependence, we additionally suggest inserting $\mathbf{\Xi} \neq \mathbf{I}$, reflecting the MA structure of the differenced data, to adjust the optimization beyond the constraint alone; in extreme cases, $d = 2$ may be warranted.

To conclude, the discrepancy between expected and empirical performance (table 4 vs. 5) could be exploited as a diagnostic tool to infer the pertinence of the model underlying $\mathbf{\Xi}$; see McElroy and Wildi (2010). We do not pursue this idea here.

	I-SSA (RW)	I-SSA (Cor)	HP-B-one
MSE	0.00010	0.00016	0.00022
HT	7.03846	10.16667	8.56250
Curvature	0.00305	0.00200	0.00235

Table 4: Empirical MSE, HT and Curvature for one-sided HP-B and I-SSA applied to log-INDPRO. The two I-SSA are based on the default RW-model but the imposed HTs differ: the first (RW) replicates the HT of HP-B when applied to white noise; the second (Cor) replicates the HT of HP-B when applied to an AR(1) with $a_1 = 0.3$. This correction affects only the specification of the HT in the constraint and does not affect optimization which is based on the plain random-walk model.

	I-SSA (RW)	I-SSA (Cor)
MSE	0.00008	0.00013
HT	5.63323	7.43516

Table 5: True (expected) MSE and HT of I-SSA based on the default RW model, with the HT in the constraint based on white noise (left) or AR(1) (right). MSE is the minimized value of the criterion. The positive bias of the empirical estimates in each HT setting suggests positive dependence in the differenced data: were the differenced INDPRO white noise, the table suggests the empirical MSE and HT would be smaller than observed.

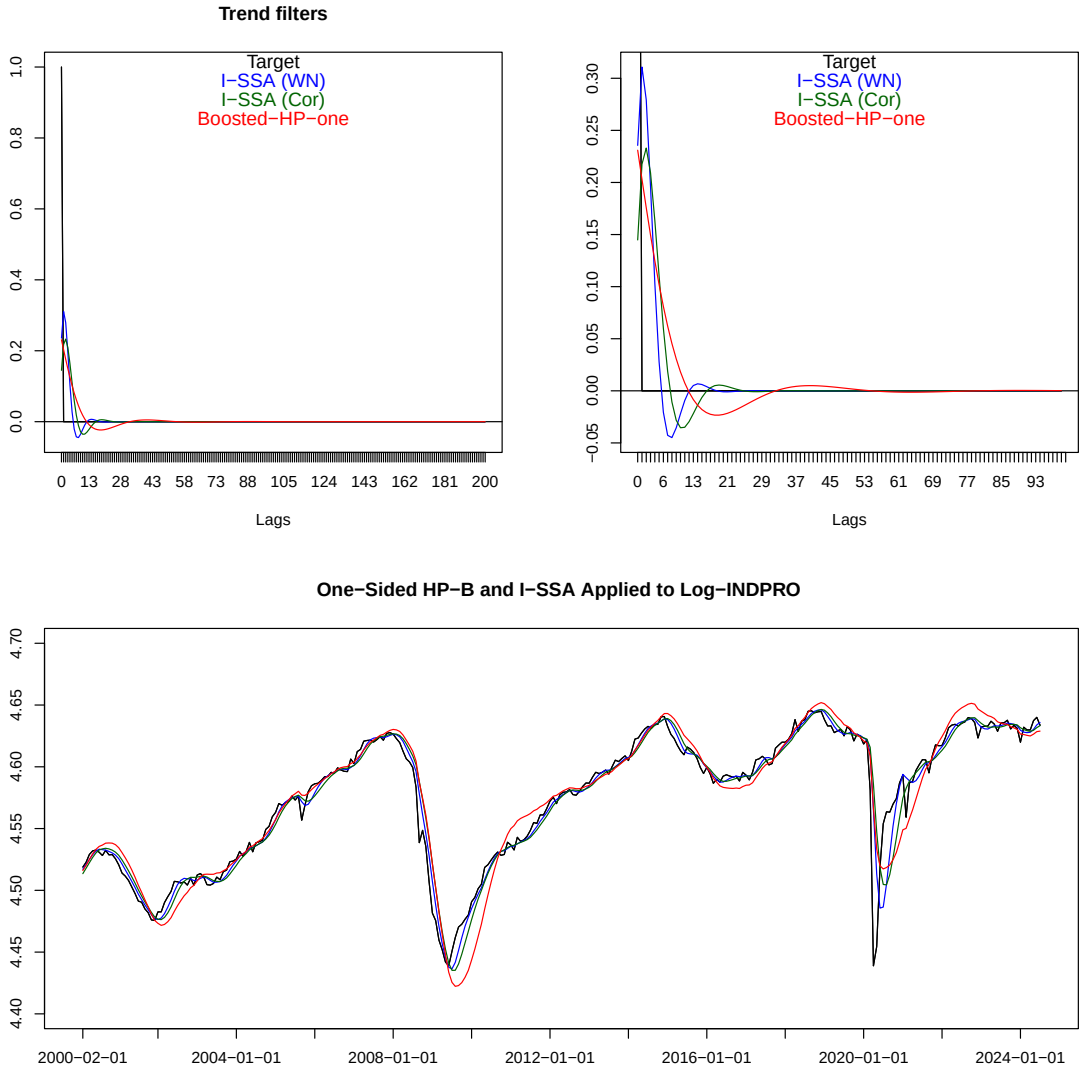


Figure 6: Log-INDPRO (black), HP-one (red), I-SSA assuming white noise (blue), and corrected I-SSA assuming an AR(1) (green) from Jan:2000-Jan:2024, covering the last three recession episodes. For similar —slightly smaller and slightly larger— TP rates, I-SSA is more adaptive and faster, with smaller retardation, tracking maxima and minima at TPs better.

5 Conclusion

I-SSA extends the SSA framework, and its underlying AS dilemma and efficient AS frontier, to non-stationary integrated time series. Its criterion defines I-SSA as the MSE-optimal predictor for a given TP/IP rate or, dually, as the maximally monotonous (for $d = 1$) or minimally flexed (for $d = 2$) predictor for a given MSE, with straightforward extensions to $d > 2$. The dilemma between the conflicting A- and S-components is thereby expressed in closed form along an efficient AS frontier.

I-SSA generalizes the classic MSE approach, a single point on this frontier, to the entire frontier. It is simple and interpretable, incorporating sign accuracy, MSE, and smoothing requirements. It identifies TPs or IPs of non-stationary series, key determinants for decision making, by optimally tracing the changing levels of a dynamically evolving phenomenon, under a neutral (amorphous-memory) smoothness constraint that is unbiased toward any specific pattern or structure.

The I-SSA solution is the global minimum of the optimization GTRS problem with hidden convexity: once the classic MSE or benchmark predictor is known, the task reduces to a one-dimensional strictly monotonic problem, solved by fast algorithms such as bisection over the strict-positivity interval with known boundaries. The problem admits an asymptotic solution, devoid of the boundary perturbation induced by the cointegration constraint. The finite sample I-SSA converges in suitable norm to the limiting I-SSA.

Under correct model specification (integration order and Wold decomposition of the stationary differences), empirical HT and MSE asymptotically match their expected values; under misspecification, they are biased. In economic applications, the default random-walk model often performs well, owing to the typical spectral shape of series accumulating noise or news. Otherwise, the approach extends readily by correcting the HT bias or by incorporating the (estimated) Wold decomposition into the problem structure, e.g., the Hessian.

A powerful use case is the customization of generic linear benchmark predictors, filters, or smoothers onto the non-stationary AS frontier, as illustrated for the HP and boosted HP filters, with further benchmarks in the I-SSA R-package. Tracking the identity, the simplest benchmark, defines the I-SSA trend, which minimizes MSE relative to the original data subject to maximal monotonicity (or minimal inflection). For matched HTs, I-SSA filters decay faster than HP or boosted HP, emphasizing adaptivity and advancement of nowcasts, with stronger side lobes accommodating the imposed HT. Tracking thus improves, particularly at key TPs (or IPs).

Looking forward, we aim to generalize I-SSA to multivariate prediction and to investigate timeliness, i.e., predictor retardation and advancement, thereby establishing a foundational prediction trilemma.

A Discussion and Interpretation of Theoretical Results

Discussion Theorem 1:

Equation (11) generalizes the spectral decomposition of the Hessian from stationary SSA (Theorem 1, Wildi, 2024 and 2026a). Unlike SSA, where $\mathbf{V}$ is the fixed Fourier eigenvector matrix of $\mathbf{M}$, here $\mathbf{V} = \mathbf{V}(\rho_1)$ is the generalized eigenvector basis of the pencil $(\mathbf{A}'\mathbf{A}, \mathbf{Q}(\rho_1))$: generally non-orthogonal and ρ_1-dependent, coinciding with the classic Fourier eigenvectors only at the boundary $\rho_1 \to \pm\rho_{\max}(L)$, $|\lambda| \to \infty$ (extremal smoothing or un-smoothing, see Appendix). The classic eigendecomposition of $\mathbf{H}(\lambda)$, by contrast, is orthogonal but depends on both ρ_1 and λ, and cannot achieve the decluttered form of (11): this requires the eigenvectors to simultaneously diagonalize the non-stationary unit-root operator $\mathbf{A}'\mathbf{A}$ (level tracking) and the stationary autocorrelation operator $\mathbf{Q}$ (smoothness)—impossible for the ordinary basis, since long memory renders these matrices non-commutable in I-SSA (see proof). The generalized decomposition is thus key to extending SSA to I-SSA subject to this structural difference tied to long memory.

Since $\mathbf{V}$ is generally non-orthogonal ($\mathbf{V}' \neq \mathbf{V}^{-1}$), the generalized spectral factorization lacks the usual traits of a classic frequency-domain representation, yet it extends stationary SSA, where

the interpretation is clear. There, the Hessian takes the simpler form $\mathbf{H}(\lambda) = \lambda\mathbf{M} + \mathbf{I}$[9], with $|\lambda| < 1/\rho_{\max}(L)$ corresponding to J_{++} in I-SSA (Wildi 2024, 2026a). Then $\mathbf{H}(\lambda)^{-1} = \mathbf{V}\mathbf{D}^{-1}\mathbf{V}'$, with Fourier eigenvectors $\mathbf{V}$ and diagonal elements $1/(1+\lambda\nu_k) = 1/(1+\lambda\cos(k/(L+1)))$, where $\nu_k = \cos(k/(L+1))$ is the k-th eigenvalue of $\mathbf{M}$, independent of ρ_1. In the time domain, the Lagrangian normal equations of SSA, $\mathbf{H}(\lambda)\mathbf{b} = D\boldsymbol{\gamma}$, read

$$0.5\lambda\Big(b_{k-1} + b_k/(0.5\lambda) + b_{k+1}\Big) = D\gamma_k, \tag{19}$$

with $\boldsymbol{\gamma}$ the MSE target and D a scaling. Its inversion $\mathbf{b} = D\mathbf{H}(\lambda)^{-1}\boldsymbol{\gamma}$ admits the frequency-domain form

$$\mathbf{b} = D\sum_{k=0}^{L}\frac{w_k}{1+\lambda\nu_k}\mathbf{v_k}, \tag{20}$$

where $w_k = \mathbf{v}_k'\boldsymbol{\gamma}$ — the special case of (11) when the generalized eigenvectors reduce to classic Fourier eigenvectors (Wildi 2024, 2026a). Rewriting

$$\frac{1}{1+\lambda\nu_k} = \frac{1}{0.5\lambda}\frac{1}{\Big(\exp(-ik/(L+1)) + 1/(0.5\lambda) + \exp(ik/(L+1))\Big)} \tag{21}$$

links both domains explicitly: $\exp(-ik/(L+1))$ and $\exp(ik/(L+1))$ are the forward and backward operators matching b_{k-1} and b_{k+1} in (19). Thus $1/(1+\lambda\nu_k)$ is the transfer function of a non-stationary, acausal, time-reversible AR(2) operator (Wildi, 2026a); $w_k/(1+\lambda\nu_k)$ is its convolution with the MSE target. This operator is lowpass for $0 > \lambda > -1$ ($\rho_1 > \rho_{MSE}$: stronger smoothing, fewer crossings), highpass for $0 < \lambda < 1$ ($\rho_1 < \rho_{MSE}$: more crossings), and allpass at $\lambda = 0$, $\rho_1 = \rho_{MSE}$ (identity). At $\lambda = \pm 1$, the stable and unstable real roots merge into a real I(2) double unit root at frequency 0 or π, respectively; beyond this, $|\lambda| > 1$, the double root bifurcates into a pair of complex conjugate unit roots at $\pm\omega \neq 0, \pi$, with the operator turning bandpass, peaking at $\pm\omega$. At $\lambda = -1/\rho_{\max}(L)$ the Hessian is semi-definite and the unit-root frequency matches the Fourier eigenvector $\mathbf{v}_{\rho_{\max}}(L)$ of $\mathbf{M}$, and similarly for $\lambda = 1/\rho_{\max}(L)$. Typically, only the unstable lowpass case (smoothing), $0 > \lambda > -1$, is relevant in applications. Instability is contained by the Fourier eigenvectors, formalized as implicit Dirichlet boundary constraints on (19) (Theorem 1, Wildi, 2024, 2026a).

With this background, and the necessary caveats, we interpret the I-SSA decomposition (11) similarly: $\frac{1}{1+\lambda\nu_i}$ extends the non-stationary SSA AR(2) operator, and its product with w_i extends the convolution with the MSE target. Note, however, that both the Hessian and ν_i now depend on ρ_1, and that the generalized spectral decompositions of I-SSA and target generally rest on different, non-orthogonal bases—what we term the 'counterfactual' spectral decomposition. The immanent instability of the AR(2) is likewise contained by the generalized eigenvectors $\mathbf{V}$, formalized by identical Dirichlet boundary constraints (Corollary 1 below), which give rise to the characteristic tipped, nose-shaped predictor profiles (see next section) observed in applications, shared by both SSA and I-SSA.

To conclude our discussion of the Theorem, a subtle but decisive difference separates SSA and I-SSA: in the stationary case, the geometry of the spectral decomposition is independent of the HT hyperparameter ρ_1, since the Fourier eigenvectors are fixed. Thus, a partially optimized solution $\mathbf{b}_\epsilon(\tilde{\lambda})$—i.e., a solution to the Lagrangian normal equations—at $\tilde{\lambda} \neq \lambda^*(\rho_1)$ is the SSA solution for some alternative $\tilde{\rho}_1 \neq \rho_1$: the Lagrange multiplier λ and the hyperparameter ρ_1 represent equivalent parametrizations of the problem and are exchangeable. This no longer holds in I-SSA, since the generalized eigenvectors $\mathbf{V} = \mathbf{V}(\rho_1)$ depend on ρ_1, affecting the entire geometry of the spectral representation: the partially optimized $\mathbf{b}_\epsilon(\tilde{\lambda})$ at $\tilde{\lambda} \neq \lambda^*(\rho_1)$ is generally not the I-SSA solution for a $\tilde{\rho}_1 \neq \rho_1$, since $\mathbf{V}(\rho_1) \neq \mathbf{V}(\tilde{\rho}_1)$. As λ and ρ_1 are no longer exchangeable parametrizations of a fixed geometric space, I-SSA solutions are obtained exclusively at $\lambda = \lambda^*(\rho_1)$. Equations (14)

[9]Here, the Hessian does not explicitly depend on ρ_1 but the dependence is implicit, since the optimal λ^* must comply with the HT constraint determined by ρ_1.

and (15) therefore single out this point—a distinction unnecessary in SSA. The I-SSA dilemma in (15) thus offers a cleaner, more fundamental formulation of the direct dependence between $J(\lambda^*(\rho_1))$ (MSE tracking) and ρ_1 (smoothness), rather than an indirect link through λ—as in Wildi (2026a,b)—which is generally inappropriate in I-SSA.

Discussion Corollary 1:

If $\boldsymbol{\Xi} = \mathbf{I}$ (random walk), $b_{\tilde{x},L} = 0$, falling back to the classic Dirichlet boundary constraint. Otherwise, $b_{\tilde{x},L} \to 0$ as $L \to \infty$, provided the (fairly general) assumptions of Theorem 2 below are satisfied. Equation (16) is the time-domain equivalent of the generalized spectral decomposition (11), which disentangles the combined permanent and transitory effects of the summation and linearized first-order ACF operators—the former absent in the SSA difference equation (19). By diagonalizing simultaneously both operators through the generalized eigenvectors (see proof), it yields a simple, interpretable extension of the SSA-AR(2) operator in the spectral domain, as formalized by (11).

The symmetry of the matrices in (16) suggests time reversibility of the I-SSA difference equation. Indeed, the first-order ACF in the HT constraint is indifferent to time orientation, so the I-SSA solution merges invertible (positive orientation: stable root) and non-invertible (negative orientation: unstable root) components of the homogeneous difference equation.[10] The implicit Dirichlet constraints at both ends control the aggregate solution, which combines forward and backward homogeneous solutions with the forward non-homogeneous solution—whose orientation is dictated by d_k and hence γ_k—thus guaranteeing boundary stability irrespective of the MSE target. The latter shapes the I-SSA predictor on the inner lags $k = 0, \ldots, L-1$, conditional on these immutable ghost boundaries. The degenerate case $\rho_1 = \rho_{MSE}$, specifically excluded by the corollary, is nevertheless obtained in the limit $\rho_1 \to \rho_{MSE}$; see the discussion ensuing the proof.

Discussion Theorem 2:

Remarks: Under fairly general assumptions, the theorem establishes convergence of the I-SSA problem structure and of its solution, together with polynomial decay of the predictor weights and polynomial convergence for large lags.

The polynomial decay justifies the finite MA approximation leading to (6). This is non-trivial: an extreme smoothness constraint $\rho_1 \to \rho_{\max}(L)$ forces $\mathbf{b}^*(L)$ towards $\mathbf{v}_{\max}(L)$, the Fourier eigenvector of the largest eigenvalue of $\mathbf{M}(L)$, whose coefficients do not decay polynomially. If instead ρ_1 is held fixed with $|\rho_1| < 1$, polynomial decay at sufficiently large lags is preserved as $L > L_0(\rho_1)$ increases. Therefore, HT constraint and objective function receive their intended meaning, at least up to negligible deviations, valid uniformly in L at the stationary points, i.e., under optimization.

The asymptotic form of I-SSA remains interpretable because it involves only 'stationary' (bounded on ℓ^2) operators, avoiding the infinite-length integration operator. In particular, the modified Hessian $K_\infty(\lambda^*_\infty)$ (18) replaces the original Hessian $\mathbf{H}$, and $\mathbf{b}^*_\infty := \mathbf{b}_\infty(\lambda^*_\infty)$ in (17) is expressed through its deviation $\mathbf{d}^*_\infty$ from the infinite-length MSE predictor. This form allows the cointegration term of I-SSA to cancel asymptotically, suppressing the delicate rank-one boundary perturbation of finite-sample predictors induced by the cointegration constraint. In other words, $\mathbf{b}^*_\infty$ satisfies the cointegration constraint intrinsically.

The theorem assumes that the MA and target weights are known. We do not address their finite-sample estimation, which is covered by the existing literature, nor its interaction with the I-SSA equation system, and leave these topics for future research. In our applications, we typically assume a known target filter, e.g., the Hodrick–Prescott (HP) filter, the boosted HP filter, or the identity filter— defining the I-SSA trend—, together with a generic random-walk specification for the underlying non-stationary series, so that $\boldsymbol{\Xi} = \mathbf{I}$. This choice complies with the theorem's assumption of known weights and, although not optimal, generally captures the dominant low-frequency dynamics of economic data reasonably well (see the examples in the next section).

Assumptions 1–2 are typically satisfied in practice, and our examples even exhibit exponential

[10] Invertibility does not refer to the causality of the predictor, which is determined solely by δ in $\tilde{z}_{t+\delta}$: for $\delta < 0$ (backcast) the predictor is acausal, otherwise (nowcast/forecast) it is causal.

decay. Assumption 3 can be checked heuristically by computing the quantities involved at several lengths L; these typically converge rapidly to finite limits satisfying the constraints (cf. Section 4). In applications, where L is fixed, this requires only that ρ_1 stay away from the extreme smoothing and un-smoothing boundaries, i.e. $|\rho_1| < \rho_{\max}(L)$ with a margin, a condition nearly always met in practice.

Convergence or polynomial decay can alternatively be derived by constraining the target $\hat{\gamma}_{MSE}$, yielding a so-called 'spectral balance' condition that extends the spectral completeness condition of Wildi (2024,2026a)— not shown. However, Assumption 3, a simple condition on the scalar $\lambda^*(L)$, is generally easier to verify experimentally.

B I(2)-SSA: Lowest Curvature

This section provides an overview of the application of the SSA to integrated processes of order two, denoted as I(2), emphasizing that our methodology is generic and can be extended to processes with higher integration orders $d > 2$ through straightforward modifications. We denote by $y_{t,MSE}$, with coefficients $\hat{\boldsymbol{\gamma}}_{MSE} = (\hat{\gamma}_{\tilde{x}\delta k})_{k=0,...,L-1}$, the MSE filter of length L. For the case where $d = 2$, the cointegration constraints, as delineated in Equation (4), can be divided into a 'level' constraint given by $\sum_{k=0}^{L-1} b_{\tilde{x}k} = \sum_{k=0}^{L-1} \hat{\gamma}_{\tilde{x}\delta k} =: \Gamma(0)$, analogous to the I(1) case, and an additional 'slope' constraint defined as $\sum_{k=1}^{L-1} k b_{\tilde{x}k} = \sum_{k=1}^{L-1} k\hat{\gamma}_{\tilde{x}\delta k} =: \dot{\Gamma}(0)$. The cancellation of the double unit-root by the error filter $\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}}$ results in a stationary process represented by $e_t = y_{t,MSE} - y_t$. This is formalized as follows:

$$\begin{aligned} y_{t,MSE} - y_t &= (\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\tilde{\mathbf{x}}_t = (\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\boldsymbol{\Sigma}^{2\,\prime}\boldsymbol{\Delta}^{2\,\prime}\tilde{\mathbf{x}}_t \\ &= (\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\boldsymbol{\Sigma}^{2\,\prime}\mathbf{x}_t. \end{aligned} \tag{22}$$

Here, the first $L-2$ entries of $\boldsymbol{\Delta}^{2\,\prime}\tilde{\mathbf{x}}_t$ are the stationary second order differences $x_t, x_{t-1}, ..., x_{t-(L-3)}$. We also substituted the last two entries $x_{t-(L-2)}, x_{t-(L-1)}$ of $\mathbf{x}_t$ to the right of the last equation for the last two non-stationary entries of $\boldsymbol{\Delta}^{2\,\prime}\tilde{\mathbf{x}}_t$ to the left due to the structure of $\boldsymbol{\Sigma}^{2\,\prime}$, whose last two columns are $(L-1-k)_{k=0,...,L-1}$ and $(L-k)_{k=0,...,L-1}$. This implies that the last two entries of $(\hat{\boldsymbol{\gamma}}_{MSE} - \mathbf{b}_{\tilde{x}})'\boldsymbol{\Sigma}^{2\,\prime}$ must vanish by the two cointegration constraints, a condition that similarly holds for $d > 2$. The underlying problem structure mirrors that of the I(1) case as discussed in Section (3.1), with the exception that $\boldsymbol{\Sigma}$ is replaced by $\boldsymbol{\Sigma}^2$ in Equation (22). Specifically, the processes situated to the right of Equation (22) exhibit stationarity, thereby enabling the formulation of a meaningful objective function (target correlation) along with HT- and length-constraints. Consequently, the optimization criteria for I(2)-SSA are derived from Equation (7), where $\boldsymbol{\Sigma}^{2\,\prime}$ replaces $\boldsymbol{\Sigma}'$:

$$\left.\begin{array}{c} \max_{\mathbf{b}_\epsilon} \mathbf{b}_\epsilon'\boldsymbol{\Sigma}^{2\,\prime}\boldsymbol{\gamma}_{\tilde{\Xi}\delta} \\ \mathbf{b}_\epsilon'\mathbf{M}\mathbf{b}_\epsilon = \rho_1\mathbf{b}_\epsilon'\mathbf{b}_\epsilon \\ \mathbf{b}_\epsilon'\boldsymbol{\Sigma}^{2\,\prime}\boldsymbol{\Sigma}^2\mathbf{b}_\epsilon = l \end{array}\right\} \quad \text{and} \quad \left.\begin{array}{c} \min_{\mathbf{b}_\epsilon} (\boldsymbol{\gamma}_{\tilde{\Xi}\delta} - \boldsymbol{\Sigma}^2\mathbf{b}_\epsilon)'(\boldsymbol{\gamma}_{\tilde{\Xi}\delta} - \boldsymbol{\Sigma}^2\mathbf{b}_\epsilon) \\ \mathbf{b}_\epsilon'\mathbf{M}\mathbf{b}_\epsilon = \rho_1\mathbf{b}_\epsilon'\mathbf{b}_\epsilon \end{array}\right\}.$$

Here, $\boldsymbol{\gamma}_{\tilde{\Xi}\delta} = \boldsymbol{\Sigma}^2\boldsymbol{\Xi}\hat{\boldsymbol{\gamma}}_{MSE}$. The HT constraint regulates the frequency of zero-crossings in the stationary second order differences $y_t - 2y_{t-1} + y_{t-2}$ of the predictor. Strict monotonicity extends straightforwardly to the I(d)-case, $d > 1$, giving raise to the efficient AS-frontier for $\rho_1 > \rho_{MSE}$ (right branch) and $\rho_1 < \rho_{MSE}$ (left branch). As a result, $y_t - 2y_{t-1} + y_{t-2}$ minimizes the number of zero-crossings subject to a specified tracking accuracy of the target variable, thus ensuring that $y_t - y_{t-1}$ exhibits maximal monotonicity. Consequently, this results in y_t having the fewest inflection points and the lowest curvature. In the final stage, it is necessary to impose level and slope (cointegration) constraints on $\mathbf{b}_{\tilde{x}}$. Solving for $b_{\tilde{x}0}$ and $b_{\tilde{x}1}$ yields $\mathbf{b}_{\tilde{x}} = (\Gamma(0) - \dot{\Gamma}(0))\mathbf{e}_1 +$

$\dot{\Gamma}(0)\mathbf{e}_2 + \mathbf{B}\tilde{\mathbf{b}}$, where $\tilde{\mathbf{b}}$ is a vector of length $L-2$, and $\mathbf{B}$ is defined as:

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & 3 & ... & L-2 \\ -2 & -3 & -4 & ... & -(L-1) \\ 1 & 0 & 0 & ... & 0 \\ 0 & 1 & 0 & ... & 0 \\ ... & & & & \\ 0 & 0 & 0 & ... & 1 \end{pmatrix}.$$

This matrix is of dimensions $L \times (L-2)$, while $\mathbf{e}_1$ and $\mathbf{e}_2$ are the first two unit-vectors of length L. An extension of Equation (9) is then straightforward.

C Proofs and Additional Results

C.1 Proofs

Proposition 1

Proof. :

By definition of $\mathbf{\Sigma}$ and $\mathbf{\Xi}$, the symmetric matrix $\mathbf{A}'\mathbf{A}$ is strictly positive definite (SPD). The matrix $\mathbf{M}$ has a decomposition with eigenvalues $\cos(j\pi/(L+1))$, $j = 1, ..., L$ —with $\rho_{\max}(L) = \cos(\pi/(L+1))$ and $\rho_{\min}(L) = -\rho_{\max}(L)$— and (Fourier) eigenvectors $\mathbf{v}_j = (\sin(jk\pi))_{k=1,...,L}$. Since $|\rho_1| < \rho_{\max}(L)$, the shifted matrix $\mathbf{M} - \rho_1\mathbf{I}$ has both strictly positive and strictly negative eigenvalues, making it indefinite, so that $\mathbf{Q} = \mathbf{\Xi}'(\mathbf{M} - \rho_1\mathbf{I})\mathbf{\Xi}$ is also symmetric and indefinite.

To determine the positive definiteness of $\mathbf{H}(\lambda)$, we analyze the inequality:

$$\mathbf{A}'\mathbf{A} + \lambda\mathbf{Q} \succ 0.$$

Since $\mathbf{A}'\mathbf{A}$ is SPD, it possesses a unique SPD square root $(\mathbf{A}'\mathbf{A})^{1/2}$. Left- and right-multiplying the inequality by $(\mathbf{A}'\mathbf{A})^{-1/2}$ preserves the positive definiteness, yielding the equivalent condition:

$$\mathbf{I} + \lambda(\mathbf{A}'\mathbf{A})^{-1/2}\mathbf{Q}(\mathbf{A}'\mathbf{A})^{-1/2} \succ 0.$$

The matrix $(\mathbf{A}'\mathbf{A})^{-1/2}\mathbf{Q}(\mathbf{A}'\mathbf{A})^{-1/2}$ is symmetric, and its eigenvalues are identical to those of $(\mathbf{A}'\mathbf{A})^{-1}\mathbf{Q}$. Let these eigenvalues be denoted by ν_i for $i = 1, \ldots, L$. Because $\mathbf{Q}$ is indefinite and $\mathbf{A}'\mathbf{A}$ is SPD, Sylvester's law of inertia dictates that $(\mathbf{A}'\mathbf{A})^{-1}\mathbf{Q}$ has both strictly positive and strictly negative eigenvalues. Thus, $\nu_{\max} > 0$ and $\nu_{\min} < 0$.

The positive definite condition requires that all eigenvalues of the shifted matrix are strictly positive:

$$1 + \lambda\nu_i > 0 \quad \text{for all } i = 1, \ldots, L.$$

This imposes two simultaneous conditions on λ: 1. For positive eigenvalues ($\nu_i > 0$), the most restrictive bound is given by $\nu_{\max}$:

$$1 + \lambda\nu_{\max} > 0 \implies \lambda > -\frac{1}{\nu_{\max}}.$$

2. For negative eigenvalues ($\nu_i < 0$), the most restrictive bound is given by $\nu_{\min}$:

$$1 + \lambda\nu_{\min} > 0 \implies \lambda < -\frac{1}{\nu_{\min}}.$$

Combining these inequalities gives the exact open interval for which $\mathbf{H}(\lambda) \succ 0$:

$$\lambda \in \left]-\frac{1}{\nu_{\max}}, -\frac{1}{\nu_{\min}}\right[.$$

Within this interval, $\mathbf{H}(\lambda)$ is strictly positive definite. □

Remark: The proof holds for arbitrary integration orders since $\mathbf{A}'\mathbf{A} = \boldsymbol{\Xi}'(\boldsymbol{\Sigma}^d)'\boldsymbol{\Sigma}^d\boldsymbol{\Xi} \succ 0$.

Theorem 1

Proof. **Part 1: Stationary Point**

First, we reparameterize the problem. Let $\mathbf{x} = \Gamma(0)\mathbf{e}_1 + \mathbf{B}\tilde{\mathbf{b}} \in \mathbb{R}^L$. By the construction of $\mathbf{B}$, the sum of its column elements is zero ($\mathbf{1}'\mathbf{B} = \mathbf{0}'$). Therefore, multiplying $\mathbf{x}$ by $\mathbf{1}'$ yields:

$$\mathbf{1}'\mathbf{x} = \Gamma(0)\mathbf{1}'\mathbf{e}_1 + \mathbf{1}'\mathbf{B}\tilde{\mathbf{b}} = \Gamma(0).$$

Optimizing over $\tilde{\mathbf{b}} \in \mathbb{R}^{L-1}$ is strictly equivalent to optimizing over $\mathbf{x} \in \mathbb{R}^L$ subject to the linear constraint $\mathbf{1}'\mathbf{x} = \Gamma(0)$. Once $\mathbf{x}$ is found, $\tilde{\mathbf{b}}$ is simply $(x_2, \ldots, x_L)'$.

Substitute $\mathbf{b}_\epsilon = \boldsymbol{\Xi}\mathbf{x}$ into the objective function:

$$J(\mathbf{x}) = (\boldsymbol{\gamma} - \boldsymbol{\Sigma}\boldsymbol{\Xi}\mathbf{x})'(\boldsymbol{\gamma} - \boldsymbol{\Sigma}\boldsymbol{\Xi}\mathbf{x}) = (\boldsymbol{\gamma} - \mathbf{A}\mathbf{x})'(\boldsymbol{\gamma} - \mathbf{A}\mathbf{x}) = \mathbf{x}'\mathbf{A}'\mathbf{A}\mathbf{x} - 2\boldsymbol{\gamma}'\mathbf{A}\mathbf{x} + \boldsymbol{\gamma}'\boldsymbol{\gamma}.$$

Substitute $\mathbf{b}_\epsilon = \boldsymbol{\Xi}\mathbf{x}$ into the quadratic constraint $\mathbf{b}_\epsilon'\mathbf{M}\mathbf{b}_\epsilon - \rho_1\mathbf{b}_\epsilon'\mathbf{b}_\epsilon = 0$:

$$\mathbf{x}'\boldsymbol{\Xi}'(\mathbf{M} - \rho_1\mathbf{I})\boldsymbol{\Xi}\mathbf{x} = \mathbf{x}'\mathbf{Q}\mathbf{x} = 0.$$

The problem is now a Quadratically Constrained Quadratic Program (QCQP) in $\mathbf{x}$:

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^L} \quad & \mathbf{x}'\mathbf{A}'\mathbf{A}\mathbf{x} - 2\boldsymbol{\gamma}'\mathbf{A}\mathbf{x} \\ \text{subject to} \quad & \mathbf{x}'\mathbf{Q}\mathbf{x} = 0, \quad \mathbf{1}'\mathbf{x} = \Gamma(0). \end{aligned}$$

We introduce Lagrange multipliers $\lambda \in \mathbb{R}$ for the quadratic constraint and $2\mu \in \mathbb{R}$ for the linear constraint. Ignoring the constant term, the Lagrangian is:

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = \mathbf{x}'\mathbf{A}'\mathbf{A}\mathbf{x} - 2\boldsymbol{\gamma}'\mathbf{A}\mathbf{x} + \lambda\mathbf{x}'\mathbf{Q}\mathbf{x} + 2\mu(\mathbf{1}'\mathbf{x} - \Gamma(0)).$$

Taking the gradient with respect to $\mathbf{x}$ and setting it to zero yields the first-order stationarity condition:

$$2\mathbf{A}'\mathbf{A}\mathbf{x} - 2\mathbf{A}'\boldsymbol{\gamma} + 2\lambda\mathbf{Q}\mathbf{x} + 2\mu\mathbf{1} = 0 \implies (\mathbf{A}'\mathbf{A} + \lambda\mathbf{Q})\mathbf{x} = \mathbf{A}'\boldsymbol{\gamma} - \mu\mathbf{1}.$$

For $\lambda \in \Lambda_{++}$, the matrix $\mathbf{H}(\lambda) = \mathbf{A}'\mathbf{A} + \lambda\mathbf{Q}$ is invertible and positive definite, ensuring a global minimum for the Lagrangian. Solving for $\mathbf{x}$ gives:

$$\mathbf{x}(\lambda, \mu) = \mathbf{H}(\lambda)^{-1}(\mathbf{A}'\boldsymbol{\gamma} - \mu\mathbf{1}).$$

To satisfy the linear constraint $\mathbf{1}'\mathbf{x} = \Gamma(0)$, we multiply by $\mathbf{1}'$:

$$\mathbf{1}'\mathbf{H}(\lambda)^{-1}\mathbf{A}'\boldsymbol{\gamma} - \mu\mathbf{1}'\mathbf{H}(\lambda)^{-1}\mathbf{1} = \Gamma(0).$$

Solving for μ provides $\mu(\lambda)$:

$$\mu(\lambda) = \frac{\mathbf{1}'\mathbf{H}(\lambda)^{-1}\mathbf{A}'\boldsymbol{\gamma} - \Gamma(0)}{\mathbf{1}'\mathbf{H}(\lambda)^{-1}\mathbf{1}}. \tag{23}$$

Substituting $\mu(\lambda)$ back into $\mathbf{x}(\lambda, \mu)$ yields $\mathbf{x}(\lambda)$ strictly as a function of the dual variable λ.

Part 2: Feasibility:

We prove the assertion by deriving a vector $\mathbf{v}^*$ that satisfies both the cointegration constraint $\mathbf{c}'\mathbf{v}^* = \Gamma(0)$, with $\mathbf{c}' = \mathbf{1}'\boldsymbol{\Xi}^{-1}$, and the HT constraint $R(\mathbf{v}^*) = \rho_1$, where $R(\mathbf{v}^*) = (\mathbf{v}^*)'\mathbf{M}\mathbf{v}^*/(\mathbf{v}^*)'\mathbf{v}^*$ denotes the Rayleigh quotient (first order ACF). Maximum $\rho_{\max}(L) = \cos(\pi/(L+1))$ and minimum $-\rho_{\max}(L) = -\cos(\pi/(L+1))$ of R are attained at $\mathbf{f}_{\max}$ and $\mathbf{f}_{\min}$, the (Fourier) eigenvectors belonging to the extremal eigenvalues $\cos(\pi/(L+1)) > \rho_1 > -\cos(\pi/(L+1))$ of $\mathbf{M}$, with the inequalities holding by assumption.

First, we find a vector $\mathbf{v}_-$ such that $R(\mathbf{v}_-) < \rho_1$ and $\mathbf{c}'\mathbf{v}_- \neq 0$. We can start with $\mathbf{f}_{\min}$. If it happens to be orthogonal to $\mathbf{c}$[11], we can perturb it infinitesimally with any vector not orthogonal to $\mathbf{c}$; by continuity of R, the perturbed vector will still have a Rayleigh quotient strictly greater than ρ_1. We do the same to find $\mathbf{v}_+$ such that $R(\mathbf{v}_+) > \rho_1$ and $\mathbf{c}'\mathbf{v}_+ \neq 0$.

Next, we normalize both vectors so they satisfy the linear constraint:

$$\mathbf{x}_+ = \frac{\Gamma(0)}{\mathbf{c}'\mathbf{v}_+}\mathbf{v}_+, \qquad \mathbf{x}_- = \frac{\Gamma(0)}{\mathbf{c}'\mathbf{v}_-}\mathbf{v}_-,$$

with $R(\mathbf{x}_+) > \rho_1$ and $R(\mathbf{x}_-) < \rho_1$, and $\mathbf{c}'\mathbf{x}_+ = \Gamma(0)$, $\mathbf{c}'\mathbf{x}_- = \Gamma(0)$. Consider the convex combination:

$$\mathbf{v}(t) = t\mathbf{x}_+ + (1-t)\mathbf{x}_-, \quad t \in [0,1].$$

For any t, the linear cointegration constraint is preserved. Since $\Gamma(0) \neq 0$, the vector $\mathbf{v}(t)$ is never the zero vector for $t \in [0,1]$, so the function $f(t) = R(\mathbf{v}(t))$ is well-defined and continuous on $[0,1]$. Since $f(1) = R(\mathbf{x}_+) > \rho_1$ and $f(0) = R(\mathbf{x}_-) < \rho_1$, the Intermediate Value Theorem yields some $t^* \in (0,1)$ with $f(t^*) = \rho_1$. Setting $\mathbf{v}^* = \mathbf{v}(t^*)$ gives a vector satisfying both $\mathbf{c}'\mathbf{v}^* = \Gamma(0)$ and $R(\mathbf{v}^*) = \rho_1$. Hence either there exists a feasible $\mathbf{b}^*$ satisfying both constraints with minimum MSE strictly smaller than that of $\mathbf{v}^*$, the I-SSA solution, or $\mathbf{b}^* = \mathbf{v}^*$ is the (feasible) I-SSA solution.

Part 3: Uniqueness:

The parametric vector $\mathbf{x}(\lambda)$ and the Lagrange multiplier $\mu(\lambda)$ satisfy the first-order optimality condition for the Lagrangian:

$$\mathbf{H}(\lambda)\mathbf{x}(\lambda) + \mu(\lambda)\mathbf{1} = \mathbf{A}'\boldsymbol{\gamma}, \tag{24}$$

subject to the linear constraint $\mathbf{1}'\mathbf{x}(\lambda) = \Gamma(0)$.

Differentiating (24) with respect to λ yields:

$$\mathbf{Q}\mathbf{x}(\lambda) + \mathbf{H}(\lambda)\dot{\mathbf{x}}(\lambda) + \dot{\mu}(\lambda)\mathbf{1} = \mathbf{0},. \tag{25}$$

Consider the secular function $F(\lambda) = \mathbf{x}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda)$, with root

$$F(\lambda^*) = 0. \tag{26}$$

The derivative of $F(\lambda)$ is:

$$\dot{F}(\lambda) = 2\dot{\mathbf{x}}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda).$$

Pre-multiplying equation (25) by $\dot{\mathbf{x}}(\lambda)'$, we obtain:

$$\dot{\mathbf{x}}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda) + \dot{\mathbf{x}}(\lambda)'\mathbf{H}(\lambda)\dot{\mathbf{x}}(\lambda) + \dot{\mu}(\lambda)\dot{\mathbf{x}}(\lambda)'\mathbf{1} = 0.$$

Since the constraint $\mathbf{1}'\mathbf{x}(\lambda) = \Gamma(0)$ holds for all $\lambda \in J_{++}$, differentiating both sides with respect to λ gives:

$$\mathbf{1}'\dot{\mathbf{x}}(\lambda) = 0.$$

Substituting this back into the previous equation eliminates the term involving $\dot{\mu}(\lambda)$ so that:

$$\dot{\mathbf{x}}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda) = -\dot{\mathbf{x}}(\lambda)'\mathbf{H}(\lambda)\dot{\mathbf{x}}(\lambda).$$

Therefore, the derivative of the secular function is:

$$\dot{F}(\lambda) = -2\dot{\mathbf{x}}(\lambda)'\mathbf{H}(\lambda)\dot{\mathbf{x}}(\lambda) < 0, \tag{27}$$

since the Hessian is strictly positive definite in J_{++}. By strict monotonicity, the root λ^* in (26) must be unique, as claimed.

[11] If $\boldsymbol{\Xi} = \mathbf{I}$ (the non-stationary process is a random-walk) and L is even, then $\mathbf{c}'\mathbf{f}_{\min} = \mathbf{1}'\mathbf{f}_{\min} = 0$.

Part 4: Global Minimum: A proof could be derived by relying on strong duality for the primal GTRS structure of I-SSA.[12] Under the posited assumptions, however, a more elementary proof is available. The Lagrangian is a quadratic function of $\mathbf{x}$, with vanishing gradient $\nabla_{\mathbf{x}}\mathcal{L}(\mathbf{x}, \lambda^*, \mu^*)|_{\mathbf{x}^*} = \mathbf{0}$, and strictly positive definite Hessian $\mathbf{H}(\lambda^*) \succ 0$ at the I-SSA solution $\mathbf{x}^* = \mathbf{x}^*(\lambda^*)$, where we assume $\mu^* = \mu(\lambda^*)$ is given by (23). Therefore $\mathbf{x}^*(\lambda^*)$ is the global minimum of $\mathcal{L}(\mathbf{x}, \lambda^*, \mu^*)$. Moreover, λ^* and $\mu(\lambda^*)$ are such that $\mathbf{x}^*$ is feasible, implying

$$\mathcal{L}(\mathbf{x}^*, \lambda^*, \mu^*) = J(\mathbf{x}^*).$$

Let $\mathbf{x}^{**}, \lambda^{**}, \mu^{**}$ be another feasible solution, so that

$$J(\mathbf{x}^{**}) = \mathcal{L}(\mathbf{x}^{**}, \lambda^{**}, \mu^{**}) = \mathcal{L}(\mathbf{x}^{**}, \lambda, \mu),$$

where equality holds for arbitrary λ, μ since the constraints vanish at any feasible solution. Then

$$J(\mathbf{x}^{**}) = \mathcal{L}(\mathbf{x}^{**}, \lambda^{**}, \mu^{**}) = \mathcal{L}(\mathbf{x}^{**}, \lambda^*, \mu^*) > \mathcal{L}(\mathbf{x}^*, \lambda^*, \mu^*) = J(\mathbf{x}^*),$$

showing that $\mathbf{x}^*$ is the strict global minimizer of MSE under HT and cointegration constraints, as claimed.

Part 5: Generalized Spectral Decomposition

Consider the generalized eigenvalue decomposition of $\mathbf{Q}$ with respect to the positive definite matrix $\mathbf{A}'\mathbf{A}$. There exists an invertible matrix $\mathbf{V}$ such that:

$$\mathbf{V}'\mathbf{A}'\mathbf{A}\mathbf{V} = \mathbf{I}, \quad \mathbf{V}'\mathbf{Q}\mathbf{V} = \mathbf{D},$$

where $\mathbf{D} = \text{diag}(\nu_1, \nu_2, \ldots, \nu_L)$ contains the generalized eigenvalues in decreasing order, so $\nu_1 = \nu_{\max}$ and $\nu_L = \nu_{\min}$. The columns of $\mathbf{V}$ are the corresponding eigenvectors $\mathbf{v}_i$.

Using this decomposition, we can rewrite $\mathbf{H}(\lambda)$ as:

$$\mathbf{H}(\lambda) = (\mathbf{V}^{-1})'(\mathbf{I} + \lambda\mathbf{D})\mathbf{V}^{-1}.$$

Its inverse is:

$$\mathbf{H}(\lambda)^{-1} = \mathbf{V}(\mathbf{I} + \lambda\mathbf{D})^{-1}\mathbf{V}'. \tag{28}$$

Therefore, we obtain:

$$\mathbf{b}_{\tilde{x}} = \mathbf{H}(\lambda)^{-1}\mathbf{g}(\lambda) = \mathbf{V}(\mathbf{I} + \lambda\mathbf{D})^{-1}\mathbf{V}'(\mathbf{V}')^{-1}\mathbf{w} = \sum_{i=1}^{L} \frac{w_i}{1 + \lambda\nu_i}\mathbf{v}_i, \tag{29}$$

where $\mathbf{w} = (w_1, ..., w_L)' = \mathbf{V}'\mathbf{g}(\lambda)$ is the 'counterfactual' spectral factorization of the target. If $\mathbf{V}$ were an orthogonal eigenvector basis, $(\mathbf{V}')^{-1} = \mathbf{V}$, and the decompositions of $\mathbf{b}_{\tilde{x}}$ and $\mathbf{g}(\lambda)$ would rely on the same 'spectral' basis, e.g., the Fourier-eigenvectors in the stationary SSA case. Here, however, the two decompositions use different (transposed) bases –$\mathbf{V}$ for $\mathbf{b}_{\tilde{x}}$ and $(\mathbf{V}')^{-1}$ for $\mathbf{g}(\lambda)$—, which we term 'counterfactual'. In any case, (29) generalizes the stationary SSA framework by diagonalizing simultaneously the permanent two-sided integration operator and the (linearized) first-order ACF operator, thereby unearthing an identical expression of the stationary decomposition, albeit in a generalized basis.

Part 6: AS Dilemma:

Let $V(\lambda)$ denote the partially optimized Lagrangian substituting the optimal $\mathbf{x}(\lambda)$ and $\mu(\lambda)$:

$$V(\lambda) = J(\lambda) + \lambda\mathbf{x}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda).$$

[12] By the S-Lemma (Polik 2007) and GTRS duality theory (Pong & Wolkowicz, 2014), the duality gap is zero, so global maximum of the dual and global minimum of the primal match. Since the former is constrained to positive (semi-) definiteness, $\lambda^* \in J_{++}$ must be the global optimum.

Note that the linear constraint $\mathbf{1}'\mathbf{x}(\lambda) - \Gamma(0) = 0$ holds for all λ, hence the term involving μ vanishes. By the Envelope Theorem, the total derivative of $V(\lambda)$ with respect to λ is equal to its partial derivative:

$$\frac{dJ(\lambda)}{d\lambda} + \mathbf{x}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda) + \lambda\frac{d}{d\lambda}\big(\mathbf{x}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda)\big) = \frac{dV(\lambda)}{d\lambda} = \frac{\partial\mathcal{L}}{\partial\lambda} = \mathbf{x}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda).$$

Therefore, we obtain

$$\frac{dJ(\lambda)}{d\lambda} = -\lambda\frac{d}{d\lambda}\big(\mathbf{x}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda)\big) = -\lambda\frac{dF(\lambda)}{d\lambda}, \tag{30}$$

yielding (12). Substituting $\mathbf{Q} = \boldsymbol{\Xi}'(\mathbf{M} - \rho_1\mathbf{I})\boldsymbol{\Xi}$ and $\mathbf{b}_\epsilon(\lambda) = \boldsymbol{\Xi}\mathbf{x}(\lambda)$, the quadratic form can be factored as:

$$\begin{aligned}\mathbf{x}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda) &= \mathbf{b}_\epsilon(\lambda)'(\mathbf{M} - \rho_1\mathbf{I})\mathbf{b}_\epsilon(\lambda)\\ &= \mathbf{b}_\epsilon(\lambda)'\mathbf{b}_\epsilon(\lambda)\left(\frac{\mathbf{b}_\epsilon(\lambda)'\mathbf{M}\mathbf{b}_\epsilon(\lambda)}{\mathbf{b}_\epsilon(\lambda)'\mathbf{b}_\epsilon(\lambda)} - \rho_1\right)\\ &= \|\mathbf{b}_\epsilon(\lambda)\|^2\big(\rho_{\mathbf{b}_\epsilon}(\lambda) - \rho_1\big).\end{aligned}$$

Thus

$$\frac{dJ(\lambda)}{d\lambda} = -\lambda\frac{d}{d\lambda}\big(\mathbf{x}(\lambda)'\mathbf{Q}\mathbf{x}(\lambda)\big) = \|\mathbf{b}_\epsilon(\lambda)\|^2\frac{d\rho_{\mathbf{b}_\epsilon}(\lambda)}{d\lambda} + \big(\rho_{\mathbf{b}_\epsilon}(\lambda) - \rho_1\big)\frac{d\|\mathbf{b}_\epsilon(\lambda)\|^2}{d\lambda}, \tag{31}$$

yielding the decomposition in (13). Equation (14) then follows from $\rho_{\mathbf{b}_\epsilon}(\lambda^*) - \rho_1 = 0$.

To verify Equation 15 we resort again to the Envelope Theorem stating

$$\frac{dJ^*(\rho_1)}{d\rho_1} = \left.\frac{\partial\mathcal{L}}{\partial\rho_1}\right|_{\lambda^*} = -\lambda^*(\mathbf{b}_\epsilon^*)'\mathbf{b}_\epsilon^*.$$

To determine the sign of λ^*, define the constraint function $C(\tilde{\mathbf{b}}) = \mathbf{b}_\epsilon'\mathbf{M}\mathbf{b}_\epsilon - \rho_1\mathbf{b}_\epsilon'\mathbf{b}_\epsilon$, where $\mathbf{b}_\epsilon$ is linked to $\tilde{\mathbf{b}}$ via (8). Here $\tilde{\mathbf{b}} \in \mathbb{R}^{L-1}$ denotes the unconstrained lower-dimensional vector in the cointegration constraint. Let $\tilde{\mathbf{b}}_{MSE}$ denote the unconstrained minimum, with associated $\mathbf{b}_{\epsilon MSE} = \boldsymbol{\Xi}\hat{\boldsymbol{\gamma}}_{MSE}$ and $J(\tilde{\mathbf{b}}_{MSE}) = 0$.

Case 1: $\rho_1 < \rho_{MSE}$. Then $C(\tilde{\mathbf{b}}_{MSE}) > 0$, and we consider the inequality constrained problem:

$$\min J(\tilde{\mathbf{b}}), \quad \text{subject to:} \quad C(\tilde{\mathbf{b}}) \leq 0,$$

where the cointegration constraint does not appear explicitly. Because the unconstrained minimum strictly violates this constraint, the constraint must be active (binding) at the optimum, meaning the solution exactly matches the equality problem $C(\tilde{\mathbf{b}}) = 0$.[13] The KKT conditions for the constraint $C(\tilde{\mathbf{b}}) \leq 0$ dictate an 'inequality' Lagrangian $\mathcal{L}_{ineq} = J(\tilde{\mathbf{b}}) + \mu C(\tilde{\mathbf{b}})$ with multiplier $\mu > 0$.[14] Matching this to our 'equality' Lagrangian, we have $\lambda^* = \mu > 0$. Therefore, $\frac{dJ^*(\rho_1)}{d\rho_1} = -\lambda^*(\mathbf{b}_\epsilon^*)'\mathbf{b}_\epsilon^* < 0$.

Case 2: $\rho_1 > \rho_{MSE}$. Then $C(\tilde{\mathbf{b}}_{MSE}) < 0$, and we consider the inequality constrained problem: $\min J(\tilde{\mathbf{b}})$ subject to $C(\tilde{\mathbf{b}}) \geq 0$, which is equivalent to $-C(\tilde{\mathbf{b}}) \leq 0$. The unconstrained minimum violates this, so the constraint is active at the optimum, matching the equality problem $C(\tilde{\mathbf{b}}) = 0$. The KKT conditions for $-C(\tilde{\mathbf{b}}) \leq 0$ dictate a Lagrangian $\mathcal{L}_{ineq} = J(\tilde{\mathbf{b}}) + \mu(-C(\tilde{\mathbf{b}}))$ with $\mu > 0$. Matching this to our equality Lagrangian, we obtain $\lambda^* = -\mu < 0$. Therefore, $\frac{dJ^*(\rho_1)}{d\rho_1} = -\lambda^*(\mathbf{b}_\epsilon^*)'\mathbf{b}_\epsilon^* > 0$, verifying (15). □

[13] Let $\tilde{\mathbf{b}}^*$ denote the optimal I-SSA solution and suppose $\tilde{\mathbf{b}}^*$ is an 'interior' point $C(\tilde{\mathbf{b}}^*) < 0$. Define $\tilde{\mathbf{b}}(t) = (1-t)\tilde{\mathbf{b}}^* + t\tilde{\mathbf{b}}_{MSE}$, $t \in [0,1]$. By convexity, $J(\tilde{\mathbf{b}}(t)) < J(\tilde{\mathbf{b}}^*)$ and by continuity we can select $\tilde{t} > 0$ such that $C(\tilde{\mathbf{b}}(\tilde{t})) < 0$; so $\tilde{\mathbf{b}}(\tilde{t})$ is feasible and with strictly smaller objective, contradicting optimality of $\tilde{\mathbf{b}}^*$.

[14] If $\mu < 0$, following the zero level-line of the constraint gradient (ensuring the constraint remains matched) would decrease the objective function, implying a maximum instead of a minimum.

Remarks: While derived for $d = 1$, the structural elements of the proof are invariant to the integration order, since $\mathbf{A}'\mathbf{A} = \boldsymbol{\Xi}'(\boldsymbol{\Sigma}^d)'\boldsymbol{\Sigma}^d\boldsymbol{\Xi} \succ 0$, so Proposition 1 holds and the Hessian can be decomposed into $\mathbf{H}(\lambda) = (\mathbf{V}^{-1})'(\mathbf{I} + \lambda\mathbf{D})\mathbf{V}^{-1}$ through the generalized eigenvalue decomposition based on the pencil $(\mathbf{A}'\mathbf{A}, \mathbf{Q})$. Only the cointegration constraint changes: none when $d = 0$ (generalizing SSA), and the adaptation in Appendix B when $d = 2$ (with straightforward extensions for $d > 2$, though largely irrelevant in applications).

The Theorem does not explicitly address the extreme HT constraint $\rho_1 = \rho_{\max}(L)$, which forces $\mathbf{b}_\epsilon = \mathbf{x}(\lambda^*) \propto \mathbf{f}_{\max}$, the Fourier eigenvector of $\mathbf{M}$ at its largest eigenvalue (and similarly $\mathbf{b}_\epsilon \propto \mathbf{f}_{\min}$ for $\rho_1 = -\rho_{\max}(L)$). Since $\mathbf{f}_{\max}$ is fixed, only sign and scaling remain free to MSE optimization.

We now briefly discuss this boundary solution via the corresponding asymptotics: as $\rho_1 \to \rho_{\max}(L)$, $\nu_{\max} \to 0^+$, $-1/\nu_{\max} \to -\infty$ (the lower bound of J_{++}), $\lambda \to -\infty$, and $\mathbf{H}(\lambda) \simeq \lambda\mathbf{Q} = \lambda\boldsymbol{\Xi}'(\mathbf{M} - \rho_{\max}(L)\mathbf{I})\boldsymbol{\Xi}$. The Hessian thus reduces to the stationary SSA case, with $\mathbf{M}$'s Fourier eigenvectors determining the spectral decomposition (Wildi, 2026a), recovering $\mathbf{b}_\epsilon \propto \mathbf{f}_{\max}$ as a limiting case. If the MSE target $\boldsymbol{\gamma}_{\tilde{\Xi}\delta} = \boldsymbol{\Sigma}\boldsymbol{\Xi}\hat{\boldsymbol{\gamma}}_{MSE}$ and the extremal I-SSA solution $\boldsymbol{\Sigma}\boldsymbol{\Xi}\mathbf{b}_{\tilde{x}} = \boldsymbol{\Sigma}\mathbf{b}_\epsilon \propto \boldsymbol{\Sigma}\mathbf{f}_{\max}$ are not mutually orthogonal, the I-SSA and MSE predictors have positive target correlation (left criterion in (7)), so the extreme HT constraint $\rho_1 = \rho_{\max}(L)$ remains compatible with the prediction problem. Otherwise, target and I-SSA are orthogonal, enforced by excessive smoothness: admissible from an optimization standpoint, but at odds with the prediction intent.

Similarly, for $\rho_1 = -\rho_{\max}(L)$, $\lim_{\lambda\to-\infty} \mathbf{b}_\epsilon(\lambda) \propto \mathbf{f}_{\min}$, the Fourier eigenvector at $-\rho_{\max}(L)$. In this extreme un-smoothing case, a further problem arises, illustrated by the random walk $\boldsymbol{\Xi} = \mathbf{I}$ with L even: $\mathbf{f}'_{\min}\mathbf{1} = 0$ implies $\mathbf{b}_{\tilde{x}}(\lambda)'\mathbf{1} = \mathbf{b}_\epsilon(\lambda)'\mathbf{1} = 0$, so $\mathbf{b}_{\tilde{x}}(\lambda)$ cancels a unit root, contradicting the standing assumption.

We therefore caution against extreme smoothing (or un-smoothing) scenarios in which $|\rho_1|$ approaches $\rho_{\max}(L)$, although in typical prediction problems ρ_1 lies well within the spectral boundaries. Moreover, increasing L to $L^+ > L$ while maintaining $|\rho_1| = \rho_{\max}(L)$ removes the singularities and releases previously frozen degrees of freedom for optimization.

To conclude, the I-SSA problem exhibits an interesting singularity cancellation as λ approaches the boundaries of J_{++}, linked to the so-called 'hard case' of the underlying GTRS problem. This cancellation becomes increasingly intricate as $|\rho_1|$ also approaches its boundary $\rho_{\max}(L)$. In particular, when $\rho_1 \to \rho_{\max}(L)$ and $\lambda \to -1/\nu_{\max} \to -\infty$, the approximation $\mathbf{H}(\lambda) \simeq \lambda\mathbf{Q}$ resolves the main (asymptotic) singularity of the exact Hessian $\mathbf{H}(\lambda) = \mathbf{A}'\mathbf{A} + \lambda\mathbf{Q}$, removing the need for an explicit cancellation in this special case. Otherwise, the required cancellation is performed 'indirectly' by the cointegration constraint, which thereby addresses two distinct instability issues at once: optimal level tracking, its original purpose, and the prevention of an asymptotically infinite predictor as the Hessian turns singular. These non-trivial interactions are further affected by L, since the cointegration term vanishes asymptotically as $L \to \infty$, assuming $\rho_1 < 1$ is fixed (see the proof of Theorem 2 below). They underscore the richness of the I-SSA criterion — its 'hidden complexity'— and are illustrated in the accompanying R package by a simple example representative of typical prediction problems.

Corollary 1

Proof. To simplify notation set $\mathbf{b} := \mathbf{b}_{\tilde{x}}$. By Theorem 1, the optimal vector $\mathbf{b} = (b_0, ..., b_{L-1}) \in \mathbb{R}^L$ satisfies the first-order condition (FOC) of the constrained optimization problem:

$$\big[\boldsymbol{\Xi}'\boldsymbol{\Sigma}'\boldsymbol{\Sigma}\boldsymbol{\Xi} + \lambda\boldsymbol{\Xi}'(\mathbf{M} - \rho_1\mathbf{I})\boldsymbol{\Xi}\big]\mathbf{b} = \boldsymbol{\Xi}'\boldsymbol{\Sigma}'\boldsymbol{\gamma} - \mu(\lambda)\mathbf{1}. \tag{32}$$

Pre-multiplying the FOC by $(\boldsymbol{\Xi}')^{-1}$ yields:

$$\big[\boldsymbol{\Sigma}'\boldsymbol{\Sigma} + \lambda(\mathbf{M} - \rho_1\mathbf{I})\big]\mathbf{v} = \mathbf{d}, \tag{33}$$

where $\mathbf{v} = \boldsymbol{\Xi}\mathbf{b}$ and $\mathbf{d} = \boldsymbol{\Sigma}'\boldsymbol{\gamma} - \mu(\lambda)(\boldsymbol{\Xi}')^{-1}\mathbf{1}$.

Equation (33) represents a linear system generating a difference equation for the elements v_k ($k = 0, \ldots, L-1$). For an interior index k ($0 < k < L-1$), the $k+1$-th row of the first-order

autocovariance matrix $\mathbf{M}$ contains 0.5 at columns k and $k+2$, and 0 elsewhere. Thus, the $k+1$-th equation of the system (33) takes the form:

$$\sum_{j=1}^{L}(\mathbf{\Sigma'\Sigma})_{k+1j}v_{j-1} + \lambda\big(0.5v_{k-1} - \rho_1 v_k + 0.5v_{k+1}\big) = d_k, \tag{34}$$

where $\lambda \neq 0$ (non-degenerate case: $\rho_1 \neq \rho_{MSE}$). This establishes the homogeneous structure of the difference equation governing the predictor coefficients.

To determine the left boundary behavior, we examine the actual first row of the system ($k = 0$):

$$\sum_{j=1}^{L}(\mathbf{\Sigma'\Sigma})_{1j}v_{j-1} + \lambda\big(- \rho_1 v_0 + 0.5v_1\big) = d_0.$$

For the general interior difference equation (34) to be universally consistent across the entire domain, its theoretical evaluation at $k = 0$

$$\sum_{j=1}^{L}(\mathbf{\Sigma'\Sigma})_{1j}v_{j-1} + \lambda\big(0.5v_{-1} - \rho_1 v_0 + 0.5v_1\big) = d_0$$

must be algebraically identical to the actual truncated first row: equality is obtained by imposing $\lambda(0.5v_{-1}) = 0$, implying the natural boundary condition:

$$v_{-1} = 0. \tag{35}$$

We now map this condition back to the original predictor coefficients $\mathbf{b}$. The relationship at the boundary is $v_k = b_k + \sum_{j=0}^{k-1}\xi_{j+1}b_{k-(j+1)}$. Extending this mapping to the ghost index $k = -1$ yields $v_{-1} = b_{-1} = 0$ ('empty' sum).

Similar reasoning holds at the right boundary. For $k = L-1$, the homogeneous structure of the difference equation is

$$\sum_{j=1}^{L}(\mathbf{\Sigma'\Sigma})_{Ljv_{j-1}} + \lambda(0.5v_{L-2} - \rho_1 v_{L-1} + 0.5v_L) = d_{L-1}.$$

It corresponds to the boundary behaviour

$$\sum_{j=1}^{L}(\mathbf{\Sigma'\Sigma})_{Ljv_{j-1}} + \lambda(0.5v_{L-2} - \rho_1 v_{L-1}) = d_{L-1},$$

implying $0.5v_L = 0$, i.e.,

$$0 = v_L = b_L + \sum_{j=0}^{L-2}\xi_{j+1}b_{L-(j+1)},$$

as claimed. □

Remarks: The proof falls back to the stationary case, because left and right boundary constraints for the 'ghosts' b_{-1} and b_L are determined entirely by $\mathbf{M}$ at $k = 0$ and $k = L-1$. The (purely causal) summation operator $\mathbf{\Sigma}$ and its reflection $\mathbf{\Sigma'}$ at the right boundary do not transgress the lag limits $k = 0, ..., L-1$.

As $\rho_1 \to \rho_{MSE}$, the Lagrange multiplier $\lambda \to 0$ and $\mathbf{b} \to \hat{\boldsymbol{\gamma}}_{MSE}$. While the two Dirichlet conditions remain active at the 'ghost' lags, the boundary adjustment term (the homogeneous solution) loses memory, confining its effect to b_{-1} and b_L. Thus, as I-SSA converges to the MSE predictor, the HT constraint loses weight through diminishing λ: the Dirichlet constraints still hold strictly but cease to influence the predictor coefficients b_k, $k = 0, \ldots, L-1$ (see Wildi 2026a and the accompanying R-package for illustration). The Corollary therefore implicitly covers the degenerate

case $\lambda = 0$, where $\mathbf{b} = \hat{\gamma}_{MSE}$ is asymptotically compatible with any values of $\hat{\gamma}_{MSE,-1}, \hat{\gamma}_{MSE,L}$, while $b_{-1} = b_L = 0$ holds strictly.

Theorem 2:

Proof. By assumption and Theorem 1, a feasible I-SSA solution $\mathbf{x}(\lambda^*(L))$ exists for all $L > L_0$ since $\rho_1 < \rho_{\max}(L_0) < \rho_{\max}(L)$.

Part (i): Convergence of the boundaries of $J_{++}(L)$.
Step 1: Elimination of $\boldsymbol{\Xi}$ and Reduction to an Eigenvalue Problem. By Sylvester's law of inertia

$$\mathbf{H}(\lambda) \succ \mathbf{0} \iff \boldsymbol{\Delta}'(\boldsymbol{\Xi}^{-1})'\mathbf{H}(\lambda)\boldsymbol{\Xi}^{-1}\boldsymbol{\Delta} = \mathbf{I} + \lambda\boldsymbol{\Delta}'(\mathbf{M} - \rho_1\mathbf{I})\boldsymbol{\Delta} \succ \mathbf{0}. \tag{36}$$

Define the symmetric matrix $\mathbf{S} := \boldsymbol{\Delta}'(\mathbf{M} - \rho_1\mathbf{I})\boldsymbol{\Delta}$, and let $\mathbf{J}$ denote the orthogonal 'anti-diagonal' identity, which satisfies $\mathbf{J}' = \mathbf{J} = \mathbf{J}^{-1}$. Then $\mathbf{J}\boldsymbol{\Delta}\mathbf{J} = \boldsymbol{\Delta}'$, $\mathbf{J}\boldsymbol{\Delta}'\mathbf{J} = \boldsymbol{\Delta}$, and $\mathbf{J}(\mathbf{M} - \rho_1\mathbf{I})\mathbf{J} = \mathbf{M} - \rho_1\mathbf{I}$, by symmetry. Consequently, $\mathbf{S}$ is orthogonally similar to

$$\mathbf{O} := \mathbf{JSJ} = \boldsymbol{\Delta}(\mathbf{M} - \rho_1\mathbf{I})\boldsymbol{\Delta}',$$

i.e., eigenvalues are identical. For all $L \geq L_0$, the assumption $|\rho_1| < \rho_{\max}(L_0) \leq \rho_{\max}(L)$ implies

$$\lambda_{\min}(\mathbf{M} - \rho_1\mathbf{I}) = -\rho_{\max}(L) - \rho_1 < 0 \quad \text{and} \quad \lambda_{\max}(\mathbf{M} - \rho_1\mathbf{I}) = \rho_{\max}(L) - \rho_1 > 0.$$

Hence, $\mathbf{M} - \rho_1\mathbf{I}$ (and consequently $\mathbf{S}$ and $\mathbf{O}$, by Sylvester's law) is strictly indefinite for all $L \geq L_0$, possessing both strictly positive and strictly negative real eigenvalues.

Let $\nu_{\max}(L) > 0$ and $\nu_{\min}(L) < 0$ denote the largest and smallest eigenvalues of $\mathbf{S}$ (equivalently of $\mathbf{O}$). Since $\mathbf{I} + \lambda\mathbf{S} \succ \mathbf{0}$ holds if and only if $1 + \lambda\nu > 0$ for all eigenvalues ν of $\mathbf{S}$, the boundaries of $J_{++}(L) = (\lambda_{\min}(L), \lambda_{\max}(L))$ are given by

$$\lambda_{\min}(L) = -\frac{1}{\nu_{\max}(L)} < 0, \qquad \lambda_{\max}(L) = -\frac{1}{\nu_{\min}(L)} > 0. \tag{37}$$

Step 2: Exact Diagonalization in the Fourier Basis and Cauchy Interlacing. Consider the identity:

$$\boldsymbol{\Delta}'\boldsymbol{\Delta} = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & -1 \\ 0 & 0 & 0 & -1 & 1 \end{pmatrix} = 2(\mathbf{I} - \mathbf{M}) - \mathbf{e}_L\mathbf{e}_L',$$

where $\mathbf{e}_L = (0, \ldots, 0, 1)'$. Define the $(L+1) \times L$ matrix obtained by appending one row to $\boldsymbol{\Delta}$,

$$\widetilde{\boldsymbol{\Delta}} := \begin{pmatrix} \boldsymbol{\Delta} \\ -\mathbf{e}_L' \end{pmatrix} \in \mathbb{R}^{(L+1)\times L}, \qquad \text{for which} \qquad \widetilde{\boldsymbol{\Delta}}'\widetilde{\boldsymbol{\Delta}} = \boldsymbol{\Delta}'\boldsymbol{\Delta} + \mathbf{e}_L\mathbf{e}_L' = 2(\mathbf{I} - \mathbf{M}) =: \mathbf{L}_D.$$

Now $\mathbf{L}_D = 2(\mathbf{I} - \mathbf{M})$ and $\mathbf{M} - \rho_1\mathbf{I}$ commute and are simultaneously diagonalized by the Fourier eigenvectors $\{\mathbf{v}^{(m)}\}_{m=1}^{L}$ of $\mathbf{M}$, with components

$$v_k^{(m)} = \sqrt{\frac{2}{L+1}}\sin(k\omega_m), \qquad \omega_m = \frac{m\pi}{L+1}, \qquad k, m \in \{1, \ldots, L\}.$$

Since $\mathbf{M}\mathbf{v}^{(m)} = \cos(\omega_m)\mathbf{v}^{(m)}$, the eigenvalues of $\mathbf{L}_D$ and $\mathbf{M} - \rho_1\mathbf{I}$ associated with mode m are $2(1 - \cos(\omega_m))$ and $(\cos(\omega_m) - \rho_1)$, respectively. Consequently, the eigenvalues of $\widetilde{\boldsymbol{\Delta}}'\widetilde{\boldsymbol{\Delta}}(\mathbf{M} - \rho_1\mathbf{I}) = \mathbf{L}_D(\mathbf{M} - \rho_1\mathbf{I}) = \mathbf{V}\mathbf{D}_{\mathbf{L}}\mathbf{V}'\mathbf{V}\mathbf{D}_{\mathbf{M}-\rho_1\mathbf{I}}\mathbf{V}' = \mathbf{V}\mathbf{D}_{\mathbf{L}}\mathbf{D}_{\mathbf{M}-\rho_1\mathbf{I}}\mathbf{V}'$ are given by

$$\mu_m(L) = 2\big(1 - \cos(\omega_m)\big)\big(\cos(\omega_m) - \rho_1\big), \qquad \omega_m = \frac{m\pi}{L+1}, \quad m = 1, \ldots, L. \tag{38}$$

Now consider the augmented $(L+1)\times(L+1)$ symmetric matrix $\widetilde{\mathbf{O}} := \widetilde{\mathbf{\Delta}}(\mathbf{M}-\rho_1\mathbf{I})\widetilde{\mathbf{\Delta}}'$. By Sylvester's determinant identity $\det(\mu\mathbf{I}_{L+1}-\mathbf{XY}) = \mu\det(\mu\mathbf{I}-\mathbf{YX})$. Setting $\mathbf{X} := \widetilde{\mathbf{\Delta}}$ and $\mathbf{Y} = (\mathbf{M}-\rho_1\mathbf{I})\widetilde{\mathbf{\Delta}}'$ shows that the eigenvalues of $\widetilde{\mathbf{O}}$ (the zeroes of $\det(\mu\mathbf{I}_{L+1}-\mathbf{XY})$) are precisely $\{\mu_m(L)\}_{m=1}^{L}\cup\{0\}$, i.e., the zeroes of $\mu\det(\mu\mathbf{I}-\mathbf{YX})$ (where the additional zero comes from the outer μ). Moreover, the block structure $\widetilde{\mathbf{\Delta}} = \left(\begin{smallmatrix}\mathbf{\Delta}\\ -\mathbf{e}_L'\end{smallmatrix}\right)$ implies that the unaugmented $L\times L$ matrix $\mathbf{O} = \mathbf{\Delta}(\mathbf{M}-\rho_1\mathbf{I})\mathbf{\Delta}'$ is the leading $L\times L$ principal submatrix of $\widetilde{\mathbf{O}}$. By Cauchy's Interlacing Theorem for symmetric matrices, the L ordered eigenvalues of $\mathbf{O}$ (and hence of $\mathbf{S}$) interlace the $L+1$ ordered eigenvalues of $\widetilde{\mathbf{O}}$. For all L sufficiently large, $\{\mu_m(L)\}_{m=1}^{L}$ in (38) contains at least two strictly negative (when $m = L-1, L$ and L large, so $\omega_m \simeq \pi$) and at least two strictly positive values ($m = 1, 2$, $\omega_m \simeq 0$), so the extra eigenvalue 0 lies strictly in the interior of the spectrum of $\widetilde{\mathbf{O}}$. Thus $\nu_{\min}(L)$ is sandwiched between the two smallest (most negative) values of $\{\mu_m(L)\}_{m=1}^{L}$, and $\nu_{\max}(L)$ is sandwiched between the two largest positive values of $\{\mu_m(L)\}_{m=1}^{L}$.

To analyze the extrema of $\{\mu_m(L)\}_{m=1}^{L}$ in (38), introduce the change of variable $c = \cos(\omega) \in (-1,1)$ and define the (quadratic) function

$$q(c) := 2(1-c)(c-\rho_1), \qquad c\in[-1,1],$$

so that $\mu_m(L) = q(\cos(\omega_m))$ for $m = 1,\dots,L$.

Step 3: Convergence of $\lambda_{\max}(L)$ as $L\to\infty$. By (37), $\lambda_{\max}(L) = -1/\nu_{\min}(L)$. Note that for $c\in[-1,1)$, $q(c) < 0$ if and only if $c\in[-1,\rho_1)$. Differentiating $q(c)$ with respect to c yields

$$q'(c) = 2(1+\rho_1-2c).$$

For all $c\in[-1,\rho_1]$, since $\rho_1 < 1$, we have $2c \le 2\rho_1 < 1+\rho_1$, which implies

$$q'(c) \ge 2(1-\rho_1) > 0, \qquad \forall c\in[-1,\rho_1].$$

Hence, $q(c)$ is strictly increasing on $[-1,\rho_1]$. Since $\cos(\omega_m)$ is strictly decreasing with respect to $m\in\{1,\dots,L\}$ and $\cos(\omega_L) < \cos(\omega_{L-1}) < \rho_1$ for all L sufficiently large, the two smallest (most negative) values of $\{q(\cos(\omega_m))\}_{m=1}^{L}$ are attained at $m = L$ and $m = L-1$. By Cauchy's Interlacing Theorem,

$$q(\cos(\omega_L)) \le \nu_{\min}(L) \le q(\cos(\omega_{L-1})) < 0.$$

As $L\to\infty$, both $\omega_L = \frac{L\pi}{L+1}\to\pi$ and $\omega_{L-1} = \frac{(L-1)\pi}{L+1}\to\pi$, so $\cos(\omega_L), \cos(\omega_{L-1})\to -1$. By continuity of $q(c)$ and the Squeeze Theorem,

$$\lim_{L\to\infty}\nu_{\min}(L) = q(-1) = 2(1-(-1))(-1-\rho_1) = -4(1+\rho_1) < 0.$$

Substituting this limit into (37) gives

$$\lim_{L\to\infty}\lambda_{\max}(L) = -\frac{1}{\lim_{L\to\infty}\nu_{\min}(L)} = \frac{1}{4(1+\rho_1)} > 0.$$

Step 4: Convergence of $\lambda_{\min}(L)$ as $L\to\infty$. By (37), $\lambda_{\min}(L) = -1/\nu_{\max}(L)$. Observe that $q(c) > 0$ if and only if $c\in(\rho_1,1)$. On the interval $[\rho_1,1]$, $q(c) = 2(1-c)(c-\rho_1)$ is a strictly concave quadratic function ($q''(c) = -4 < 0$) vanishing at the endpoints $c = \rho_1$ and $c = 1$.

Setting $q'(c) = 2(1+\rho_1-2c) = 0$, we find that $q(c)$ attains its unique global maximum on $[-1,1]$ at the midpoint

$$c_{\max} := \frac{1+\rho_1}{2}\in(\rho_1,1),$$

corresponding to the interior frequency $\omega_{\max} := \arccos\left(\frac{1+\rho_1}{2}\right)\in(0,\arccos(\rho_1))\subset(0,\pi)$. Evaluating $q(c)$ at $c = c_{\max}$ yields

$$\max_{c\in[-1,1]} q(c) = q(c_{\max}) = 2\left(1-\frac{1+\rho_1}{2}\right)\left(\frac{1+\rho_1}{2}-\rho_1\right) = 2\left(\frac{1-\rho_1}{2}\right)^2 = \frac{(1-\rho_1)^2}{2} > 0.$$

Since the discrete grid points $\{\omega_m\}_{m=1}^{L} = \{\frac{m\pi}{L+1}\}_{m=1}^{L}$ form a uniform mesh of $(0,\pi)$ with step size $\frac{\pi}{L+1} \to 0$ as $L \to \infty$, for every L sufficiently large there exist two distinct indices $m_1(L), m_2(L) \in \{1,\dots,L\}$ such that $|\omega_{m_j(L)} - \omega_{\max}| \le \frac{\pi}{L+1}$ and $q(\cos(\omega_{m_j(L)})) > 0$ for $j = 1,2$. By Cauchy's Interlacing Theorem and the global optimality of $c_{\max}$, $\nu_{\max}(L)$ is bounded between the second-largest and the largest values of $\{q(\cos(\omega_m))\}_{m=1}^{L}$, which implies

$$\min_{j\in\{1,2\}} q\big(\cos(\omega_{m_j(L)})\big) \le \nu_{\max}(L) \le \max_{1\le m\le L} q(\cos(\omega_m)) \le q(c_{\max}).$$

Since $\omega_{m_j(L)} \to \omega_{\max}$ as $L \to \infty$ for $j = 1,2$ and $\omega \mapsto q(\cos(\omega))$ is continuous, the Squeeze Theorem yields

$$\lim_{L\to\infty} \nu_{\max}(L) = q(c_{\max}) = \frac{(1-\rho_1)^2}{2} > 0.$$

Substituting this limit into (37) gives

$$\lim_{L\to\infty} \lambda_{\min}(L) = -\frac{1}{\lim_{L\to\infty} \nu_{\max}(L)} = -\frac{2}{(1-\rho_1)^2} < 0,$$

which completes the proof of Claim (i).

Part (ii): polynomial decay of I-SSA weights.

Preamble: Uniform strict positivity. Let

$$\mathbf{K} = \mathbf{K}(\lambda^*(L)) := \mathbf{I} + \lambda^*(L)\,\mathbf{S}, \qquad \mathbf{S} = \mathbf{\Delta}'(\mathbf{M} - \rho_1\mathbf{I})\mathbf{\Delta},$$

which, by the congruence of Part (i), satisfies

$$\mathbf{K} = \mathbf{\Delta}'(\mathbf{\Xi}')^{-1}\,\mathbf{H}(\lambda^*)\,\mathbf{\Xi}^{-1}\mathbf{\Delta} \quad\Longleftrightarrow\quad \mathbf{H}^{-1} = \mathbf{\Xi}^{-1}\mathbf{\Delta}\,\mathbf{K}^{-1}\mathbf{\Delta}'(\mathbf{\Xi}')^{-1}. \tag{39}$$

By Part (i), the eigenvalues of $\mathbf{S}$ lie in $[\nu_{\min}(\mathbf{S}), \nu_{\max}(\mathbf{S})]$ with $\nu_{\min}(\mathbf{S}) \to S_{\min} < 0$ and $\nu_{\max}(\mathbf{S}) \to S_{\max} > 0$, and the boundaries $\lambda_{\min}(L), \lambda_{\max}(L)$ of $J_{++}(L)$ converge to finite limits. Hence $\Lambda := \sup|\lambda^*(L)| < \infty$. Assumption (A3) gives $\delta > 0$ with $\operatorname{dist}(\lambda^*(L), \partial J_{++}(L)) \ge \delta$ for all L, so that

$$1 + \lambda^*\nu_j \;\ge\; c_\delta > 0 \quad \text{for every eigenvalue } \nu_j \text{ of } \mathbf{S},$$

with c_δ depending only on δ and on the limits in Part (i) (for $\lambda^* > 0$: $1 + \lambda^*\nu_{\min} = \nu_{\min}(\lambda^* - \lambda_{\max}(L)) \ge |\nu_{\min}|\delta$, and analogously for $\lambda^* < 0$). Consequently there exist universal constants $0 < c \le C < \infty$, independent of L, such that

$$c\,\mathbf{I} \preceq \mathbf{K} \preceq C\,\mathbf{I} \qquad \text{uniformly in } L. \tag{40}$$

Notation and auxiliary facts. For $s > 1$ and $a > 0$, $\mathbf{N} \in \mathcal{J}_s(a)$ if $|N_{ij}| \le a(1+|i-j|)^{-s}$.

(F1) Convolution inequality. For $s > 1$ there is $c_1(s) < \infty$ with

$$\sum_{l\in\mathbb{Z}} (1+|i-l|)^{-s}(1+|l-j|)^{-s} \le c_1(s)(1+|i-j|)^{-s}.$$

Consequences (independent of L): (i) $\mathbf{N}_1 \in \mathcal{J}_s(a_1)$, $\mathbf{N}_2 \in \mathcal{J}_s(a_2) \Rightarrow \mathbf{N}_1\mathbf{N}_2 \in \mathcal{J}_s(c_1(s)a_1a_2)$; (ii) $\mathbf{N} \in \mathcal{J}_s(a)$, $|v_l| \le b(1+l)^{-s} \Rightarrow |(\mathbf{N}\mathbf{v})_k| \le c_1(s)ab(1+k)^{-s}$; (iii) a banded matrix with bandwidth β and entries bounded by m lies in $\mathcal{J}_s(m(1+\beta)^s)$.

(F2) Finite-L Jaffard theorem. If $\mathbf{K} \in \mathbb{R}^{L\times L}$, $|K_{ij}| \le a(1+|i-j|)^{-s}$ and $c\mathbf{I} \preceq \mathbf{K} \preceq C\mathbf{I}$, then $|(\mathbf{K}^{-1})_{ij}| \le a'(1+|i-j|)^{-s}$ with $a' = a'(s,a,c,C)$ (quantitative Jaffard theorem, Gröchenig–Klotz (2010)).

(F3) Decay of the inverse MA filter. Under (A1) ($|\xi_j| \le C_\xi(1+j)^{-s}$, $\xi(e^{i\omega}) \ne 0$ on $[-\pi,\pi]$ with $|\xi(e^{i\omega})| \ge m_\xi > 0$), the coefficients π_j of $1/\xi(z)$ satisfy $|\pi_j| \le C_\pi(1+j)^{-s}$, with $C_\pi = C_\pi(s, C_\xi, m_\xi)$

(Wiener Lemma, Gröchenig (2010)). Since $\boldsymbol{\Xi}_L^{-1}$ is the lower triangular Toeplitz matrix with entries π_{i-j}, we obtain $\boldsymbol{\Xi}^{-1} \in \mathcal{J}_s(C_\pi)$ uniformly in L.

Step 0: exact algebra. Since $\boldsymbol{\Delta}, \boldsymbol{\Sigma}, \boldsymbol{\Xi}$ are lower triangular Toeplitz matrices, they commute and $\mathbf{A}\boldsymbol{\Delta} = \boldsymbol{\Sigma}\boldsymbol{\Xi}\boldsymbol{\Delta} = \boldsymbol{\Xi}$. From $\boldsymbol{\gamma} = \mathbf{A}\boldsymbol{\gamma}_{MSE}$ we get $\mathbf{A}'\boldsymbol{\gamma} = \mathbf{A}'\mathbf{A}\boldsymbol{\gamma}_{MSE}$, and $\mathbf{A}'\mathbf{A} = \mathbf{H} - \lambda^*\mathbf{Q}$. Therefore, from $\mathbf{H}\mathbf{x}^* = \mathbf{A}'\boldsymbol{\gamma} - \mu\mathbf{1}$,

$$\mathbf{d} := \mathbf{x}^* - \boldsymbol{\gamma}_{MSE} = -\mathbf{H}^{-1}\big(\lambda^*\mathbf{Q}\boldsymbol{\gamma}_{MSE} + \mu\mathbf{1}\big). \tag{41}$$

Using (39) and $\mathbf{Q} = \boldsymbol{\Xi}'(\mathbf{M} - \rho_1\mathbf{I})\boldsymbol{\Xi}$,

$$\begin{aligned} \boldsymbol{\Delta}'(\boldsymbol{\Xi}')^{-1}\mathbf{Q}\boldsymbol{\gamma}_{MSE} &= \boldsymbol{\Delta}'(\mathbf{M} - \rho_1\mathbf{I})\boldsymbol{\Xi}\boldsymbol{\gamma}_{MSE} =: \mathbf{f} \\ \boldsymbol{\Delta}'(\boldsymbol{\Xi}')^{-1}\mathbf{1} &= (\boldsymbol{\Xi}')^{-1}\boldsymbol{\Delta}'\mathbf{1} = (\boldsymbol{\Xi}')^{-1}\mathbf{e}_L =: \boldsymbol{\zeta}, \end{aligned} \tag{42}$$

where $\boldsymbol{\zeta} = (\pi_{L-1}, \ldots, \pi_1, \pi_0)'$ with $\pi_0 = 1/\xi_0$. Hence

$$\mathbf{d} = -\boldsymbol{\Xi}^{-1}\boldsymbol{\Delta}\,\mathbf{w}, \qquad \mathbf{w} := \mathbf{K}^{-1}\big(\lambda^*\mathbf{f} + \mu\,\boldsymbol{\zeta}\big). \tag{43}$$

The definition of $\mu(\lambda^*)$ is equivalent to $\mathbf{1}'\mathbf{x}^* = \Gamma(0)$, and $\mathbf{1}'\boldsymbol{\gamma}_{MSE} = \Gamma(0)$, so $\mathbf{1}'\mathbf{d} = 0$, by defintion of $\mathbf{d}$ in (41). Because $\mathbf{1}'\boldsymbol{\Xi}^{-1}\boldsymbol{\Delta} = \big(\boldsymbol{\Delta}'(\boldsymbol{\Xi}')^{-1}\mathbf{1}\big)' = \boldsymbol{\zeta}'$,

$$0 = \mathbf{1}'\mathbf{d} = -\boldsymbol{\zeta}'\mathbf{w} \quad\Longleftrightarrow\quad \lambda^*\,\boldsymbol{\zeta}'\mathbf{K}^{-1}\mathbf{f} + \mu\,\boldsymbol{\zeta}'\mathbf{K}^{-1}\boldsymbol{\zeta} = 0 \quad\Longleftrightarrow\quad \mu = -\lambda^*\frac{\boldsymbol{\zeta}'\mathbf{K}^{-1}\mathbf{f}}{\boldsymbol{\zeta}'\mathbf{K}^{-1}\boldsymbol{\zeta}}, \tag{44}$$

where, by (40), $\boldsymbol{\zeta}'\mathbf{K}^{-1}\boldsymbol{\zeta} \geq \|\boldsymbol{\zeta}\|^2/C \geq \pi_0^2/C = 1/(C\xi_0^2) > 0$. In summary, step 0 isolates the difference vector $\mathbf{d} = \mathbf{x}^* - \boldsymbol{\gamma}_{MSE}$; the substitution of the congruence transformation $\mathbf{H}^{-1} = \boldsymbol{\Xi}^{-1}\boldsymbol{\Delta}\mathbf{K}^{-1}\boldsymbol{\Delta}'(\boldsymbol{\Xi}')^{-1}$ maps the problem to the well-conditioned matrix $\mathbf{K}$; identifying that $\boldsymbol{\Delta}'\mathbf{1} = \mathbf{e}_L$ localizes the linear constraint's influence to the sequence $\boldsymbol{\zeta}$, crucial for later bounding μ, noting that $\boldsymbol{\zeta}$ and $\mathbf{f}$ have time reverted polynomial decays in the numerator $\boldsymbol{\zeta}'\mathbf{K}^{-1}\mathbf{f}$ of the right-hand side quotient in (44).

Step 1: structure of K. $\mathbf{S} = \boldsymbol{\Delta}'(\mathbf{M} - \rho_1\mathbf{I})\boldsymbol{\Delta}$ is a product of banded matrices ($\boldsymbol{\Delta}$ bidiagonal, $\mathbf{M} - \rho_1\mathbf{I}$ tridiagonal), hence pentadiagonal with entries bounded by $m_S := 4\max(0.5, |\rho_1|)$. Therefore $\mathbf{K} = \mathbf{I} + \lambda^*\mathbf{S}$ is pentadiagonal with entries bounded by $1 + \Lambda m_S$, and by (F1, iii), $\mathbf{K} \in \mathcal{J}_s(a_K)$ with $a_K = (1 + \Lambda m_S)\,3^s$ (bandwidth $\beta = 2$), independent of L.

Step 2: decay of $\mathbf{K}^{-1}$. By (40), Step 1 and (F2), $\mathbf{K}^{-1} \in \mathcal{J}_s(a_{K^{-1}})$ with $a_{K^{-1}} = a'(s, a_K, c, C)$ independent of L. In summary, steps 1 and 2 establish that $\mathbf{K}^{-1}$ belongs to the Jaffard class $\mathcal{J}_s$, preserving the polynomial decay rate s of its entries uniformly in L.

Step 3: the forcing f and the vector $\boldsymbol{\zeta}$. First, $\boldsymbol{\Xi} \in \mathcal{J}_s(C_\xi)$ by (A1). By (A2) and (ii), $|(\boldsymbol{\Xi}\boldsymbol{\gamma}_{MSE})_k| \leq c_1(s)C_\xi C_\gamma(1+k)^{-s}$, $k = 1, \ldots, L$. The matrix $\boldsymbol{\Delta}'(\mathbf{M} - \rho_1\mathbf{I})$ is banded with entries bounded by $2\max(0.5, |\rho_1|) \cdot 2$, hence belongs to $\mathcal{J}_s(a_B)$ by F1 (iii). Then, from the defintion of $\mathbf{f}$ in (42) and by F1 (ii)

$$|f_k| \leq a_f(1+k)^{-s}, \qquad k = 1, \ldots, L, \qquad a_f := c_1(s)a_B c_1(s)C_\xi C_\gamma.$$

Next, by (F3), $|\zeta_j| = |\pi_{L-j}| \leq C_\pi(1 + L - j)^{-s}$ for $j = 1, \ldots, L$; in particular $\boldsymbol{\zeta}$ is concentrated near the last index.

Step 4: bounds on μ and $\mathbf{w}$. By Steps 2, 3 and (ii),

$$|(\mathbf{K}^{-1}\mathbf{f})_j| \leq c_2(1+j)^{-s}, \qquad j = 1, \ldots, L, \qquad c_2 := c_1(s)a_{K^{-1}}a_f. \tag{45}$$

Since $\boldsymbol{\zeta}'\mathbf{K}^{-1}\mathbf{f} = \sum_j \zeta_j(\mathbf{K}^{-1}\mathbf{f})_j$,

$$|\boldsymbol{\zeta}'\mathbf{K}^{-1}\mathbf{f}| \leq C_\pi c_2 \sum_{j=1}^{L}(1 + L - j)^{-s}(1+j)^{-s} \leq C_\pi c_2 c_1(s)(1+L)^{-s},$$

by (F1) (with $i = L$ after the shift $j \mapsto L - j$ and $l = 0$). Using (44) and the lower bound $\boldsymbol{\zeta}'\mathbf{K}^{-1}\boldsymbol{\zeta} \geq 1/(C\xi_0^2)$,

$$|\mu| \leq \Lambda\, C\xi_0^2\, C_\pi c_1(s) c_2\, (1+L)^{-s} =: c_\mu (1+L)^{-s}. \tag{46}$$

Furthermore, $\mathbf{K}^{-1}\boldsymbol{\zeta}$ decays away from the last index: by Step 2, (F1) and $|\zeta_l| \leq C_\pi(1+L-l)^{-s}$,

$$|(\mathbf{K}^{-1}\boldsymbol{\zeta})_j| \leq c_1(s) a_{K^{-1}} C_\pi (1+L-j)^{-s} \leq c_1(s) a_{K^{-1}} C_\pi, \qquad j = 1, \ldots, L.$$

Hence, for $j = 1, \ldots, L$, using (45), (46) and $(1+L)^{-s} \leq (1+j)^{-s}$,

$$|w_j| \leq |\lambda^*||(\mathbf{K}^{-1}\mathbf{f})_j| + |\mu||(\mathbf{K}^{-1}\boldsymbol{\zeta})_j| \leq \Lambda c_2 (1+j)^{-s} + c_\mu c_1(s) a_{K^{-1}} C_\pi (1+L)^{-s} \leq c_w (1+j)^{-s},$$

with $c_w := \Lambda c_2 + c_\mu c_1(s) a_{K^{-1}} C_\pi$, independent of L. Set $w_0 := 0$. In summary steps 3 and 4 establish that $\mathbf{f}$ decays algebraically starting from the first index $k = 1$; the vector $\boldsymbol{\zeta}$, defined by the inverse MA filter coefficients, decays algebraically starting from the last index L (i.e., it is concentrated at the right boundary); when evaluating the inner product $\boldsymbol{\zeta}'\mathbf{K}^{-1}\mathbf{f}$ to solve for the Lagrange multiplier μ, the convolution of these two opposing decays over a sequence of length L yields a uniform bound of $O((1+L)^{-s})$. This guarantees that the boundary effect induced by the linear constraint (represented by μ) vanishes polynomially as $L \to \infty$, preventing it from affecting the decay of the early weights.

Step 5: conclusion. From (43), $\mathbf{d} = -\boldsymbol{\Xi}^{-1}\mathbf{u}$ with $\mathbf{u} := \boldsymbol{\Delta}\mathbf{w}$, $u_k = w_k - w_{k-1}$, $k = 1, \ldots, L$. As in the previous argument, $|u_1| = |w_1| \leq c_w 2^{-s}$ and for $k \geq 2$

$$|u_k| \leq |w_k| + |w_{k-1}| \leq c_w(1+k)^{-s} + c_w k^{-s} \leq c_w(1+2^s)(1+k)^{-s},$$

since $k^{-s} \leq 2^s(1+k)^{-s}$ for $k \geq 1$. So $|u_k| \leq c_u(1+k)^{-s}$ with $c_u := c_w(1+2^s)$. Finally, by (F3) and (ii),

$$|d_k| = \left|(\boldsymbol{\Xi}^{-1}\mathbf{u})_k\right| \leq c_1(s) C_\pi c_u (1+k)^{-s} =: C^*(1+k)^{-s}, \qquad k = 1, \ldots, L,$$

where C^* depends only on $(s, \Lambda, c, C, C_\xi, m_\xi, C_\gamma, \rho_1, \xi_0)$ and the separation constant in (A3), but not on L. Together with (A2),

$$|x_k^*| \leq |\gamma_{MSE,k}| + |d_k| \leq (C_\gamma + C^*)(1+k)^{-s}, \qquad k = 1, \ldots, L.$$

Since $\tilde{\mathbf{b}}^* = (x_2^*, \ldots, x_L^*)'$ is a sub-vector of $\mathbf{x}^*$, the same bound holds for its components.

Part (iii): Convergence of $\lambda^*(L)$ and asymptotic I-SSA.
Recall from Step 0 of Part (ii) that, with $\mathbf{d}_L(\lambda) := \mathbf{x}_L(\lambda) - \boldsymbol{\gamma}_{MSE}$,

$$\mathbf{d}_L(\lambda) = -\boldsymbol{\Xi}_L^{-1}\boldsymbol{\Delta}_L\mathbf{w}_L(\lambda),\ \mathbf{w}_L(\lambda) = \mathbf{K}_L(\lambda)^{-1}\big(\lambda\,\mathbf{f}_L + \mu_L(\lambda)\,\boldsymbol{\zeta}_L\big),\ \mathbf{f}_L = \boldsymbol{\Delta}_L'(\mathbf{M}_L - \rho_1\mathbf{I}_L)\boldsymbol{\Xi}_L\boldsymbol{\gamma}_{MSE}, \tag{47}$$

where

$$\mathbf{K}_L(\lambda) = \boldsymbol{\Delta}_L'(\boldsymbol{\Xi}_L')^{-1}\mathbf{H}_L(\lambda)\boldsymbol{\Xi}_L^{-1}\boldsymbol{\Delta}_L = \mathbf{I}_L + \lambda\mathbf{S}_L, \qquad \mathbf{S}_L = \boldsymbol{\Delta}_L'(\mathbf{M}_L - \rho_1\mathbf{I}_L)\boldsymbol{\Delta}_L,$$

$\boldsymbol{\zeta}_L = (\boldsymbol{\Xi}_L')^{-1}\mathbf{e}_L = (\pi_{L-1}, \ldots, \pi_1, \pi_0)'$, and $\mu_L(\lambda)$ is fixed by the linear constraint, which reads $\mathbf{1}'\mathbf{d}_L = -\boldsymbol{\zeta}_L'\mathbf{w}_L = 0$, i.e. $\mu_L(\lambda) = -\lambda\,\boldsymbol{\zeta}_L'\mathbf{K}_L^{-1}\mathbf{f}_L/(\boldsymbol{\zeta}_L'\mathbf{K}_L^{-1}\boldsymbol{\zeta}_L)$. Working with $\mathbf{d}$ rather than $\mathbf{x}$ has the advantage that the forcing term involves only $\mathbf{f}_L$, which is built from $\boldsymbol{\gamma}_{MSE}$ and $\boldsymbol{\xi}$, both of which decay polynomially.

Step 1: Compactness. By (A3), $\operatorname{dist}(\lambda^*(L), \partial J_{++}(L)) \geq \delta > 0$ for all L. Since the boundaries converge to finite limits by Part (i), the sequence $\lambda^*(L)$ is bounded. Hence there is a subsequence with

$$\lim_{n\to\infty} \lambda^*(L_n) = \lambda^\circ \in [\lambda_{\min}(\infty) + \delta,\ \lambda_{\max}(\infty) - \delta] \subset J_{++}(\infty). \tag{48}$$

Step 2: Convergence of the inverse. Let $\boldsymbol{\Xi}_\infty$, $\boldsymbol{\Delta}_\infty$, $\mathbf{S}_\infty$ be the (bounded) one-sided infinite versions of $\boldsymbol{\Xi}_L$, $\boldsymbol{\Delta}_L$, $\mathbf{S}_L$; $\boldsymbol{\Xi}_\infty$ is bounded on ℓ^2 because $\boldsymbol{\xi} \in \ell^1$ by (A1), and $\mathbf{S}_\infty$ is banded. Since

λ° is strictly inside $J_{++}(\infty)$, that is, $\mathbf{I} + \lambda^\circ \mathbf{S}_\infty \succeq c_\delta \mathbf{I}$ with $c_\delta > 0$, the operator $\mathbf{K}_\infty(\lambda^\circ) = \mathbf{I} + \lambda^\circ \mathbf{S}_\infty$ is bounded and bounded below, hence invertible on ℓ^2. By the Preamble of Part (ii), $c\mathbf{I} \preceq \mathbf{K}_L(\lambda^*(L)) \preceq C\mathbf{I}$ uniformly in L. Moreover $\|\mathbf{K}_{L_n}(\lambda^*_{L_n}) - \mathbf{K}_{L_n}(\lambda^\circ)\| \le |\lambda^*_{L_n} - \lambda^\circ| \, \|\mathbf{S}_{L_n}\| \to 0$ as $\lambda^*_{L_n} \to \lambda^\circ$, uniformly in L_n (recall $\|\mathbf{S}_L\| \le 4(1 + |\rho_1|)$), and $\mathbf{K}_{L_n}(\lambda^\circ)$ converges strongly to $\mathbf{K}_\infty(\lambda^\circ)$ (entrywise convergence of uniformly bounded, uniformly banded matrices). Together with the uniform bound on the inverses, this gives

$$\mathbf{K}_{L_n}(\lambda^*(L_n))^{-1} \xrightarrow{s} \mathbf{K}_\infty(\lambda^\circ)^{-1}. \tag{49}$$

Step 3: Convergence of the forcing term. By (A1)–(A2) and Step 3 of Part (ii), $|(\mathbf{f}_L)_k| \le a_f(1+k)^{-s}$ for all L and $k \le L$, so $\mathbf{f}_L \to \mathbf{f}_\infty := \boldsymbol{\Delta}'_\infty(\mathbf{M}_\infty - \rho_1\mathbf{I})\boldsymbol{\Xi}_\infty\boldsymbol{\gamma}_{MSE}$ in ℓ^2 (dominated convergence, since $s > 1$). By (46), $|\mu_L(\lambda^*(L))| \le c_\mu(1+L)^{-s}$, and $\boldsymbol{\zeta}_L$ has bounded ℓ^2-norm, $\|\boldsymbol{\zeta}_L\|_2^2 = \sum_{j=0}^{L-1} \pi_j^2 \le C_\pi^2 \sum_{j\ge 0}(1+j)^{-2s} < \infty$, so $\mu_L(\lambda^*(L))\,\boldsymbol{\zeta}_L$ is $O(L^{-s})$ in ℓ^2 and tends to 0. Hence

$$\lambda^*(L_n)\mathbf{f}_{L_n} + \mu_{L_n}\boldsymbol{\zeta}_{L_n} \;\longrightarrow\; \lambda^\circ \mathbf{f}_\infty \quad \text{in } \ell^2, \tag{50}$$

and therefore

$$\mathbf{w}_{L_n} \to \mathbf{w}_\infty := \lambda^\circ \, \mathbf{K}_\infty(\lambda^\circ)^{-1}\mathbf{f}_\infty, \qquad \mathbf{d}_{L_n} = -\boldsymbol{\Xi}^{-1}\boldsymbol{\Delta}\mathbf{w}_{L_n} \to \mathbf{d}_\infty := -\boldsymbol{\Xi}_\infty^{-1}\boldsymbol{\Delta}_\infty\mathbf{w}_\infty \quad \text{in } \ell^2, \tag{51}$$

because $\boldsymbol{\Delta}$ and $\boldsymbol{\Xi}^{-1}$ (whose entries π_j are in ℓ^1 by (F3)) are bounded. The limit $\mathbf{d}_\infty$ automatically satisfies the linear constraint, since $\mathbf{1}'\mathbf{d}_L = -\boldsymbol{\zeta}'_L\mathbf{w}_L = 0$ for every L and the convergence $\mathbf{d}_{L_n} \to \mathbf{d}_\infty$ holds in ℓ^1 (by the uniform polynomial decay $|d_{L,k}| \le C^*(1+k)^{-s}$ of Part (ii) and dominated convergence); that is, the constraint does not need to be imposed separately in the limit.

Step 4: Limit of the stationary point. By Steps 2 and 3, together with the uniform bound on $\|\mathbf{K}_{L_n}^{-1}\|$ and the strong convergence of $\mathbf{K}_{L_n}(\lambda^*(L_n))^{-1}$, the deviation of the finite-dimensional stationary point from the MSE filter, $\mathbf{d}^*(L_n) = \mathbf{x}^*(L_n) - \boldsymbol{\gamma}_{MSE} = -\boldsymbol{\Xi}^{-1}\boldsymbol{\Delta}\,\mathbf{w}_{L_n}$, converges in ℓ^2 to

$$\mathbf{d}^*(\infty) = -\boldsymbol{\Xi}_\infty^{-1}\boldsymbol{\Delta}_\infty \, \lambda^\circ \, \mathbf{K}_\infty(\lambda^\circ)^{-1}\mathbf{f}_\infty, \tag{52}$$

so that $\mathbf{x}^*(\infty) := \boldsymbol{\gamma}_{MSE} + \mathbf{d}^*(\infty)$ is the limit of the stationary points (in the subsequence defined by L_n). It satisfies the linear constraint $\mathbf{1}'\mathbf{d}^*(\infty) = 0$ (Step 3). Passing to the limit in the stationarity condition and in the quadratic constraint $\mathbf{x}^*(L_n)'\mathbf{Q}_{L_n}\mathbf{x}^*(L_n) = 0$, equivalently $(\boldsymbol{\Xi}_{L_n}\mathbf{x}^*(L_n))'(\mathbf{M}_{L_n} - \rho_1\mathbf{I}_{L_n})(\boldsymbol{\Xi}_{L_n}\mathbf{x}^*(L_n)) = 0$ (all terms converge, since $\mathbf{x}^*(L_n)$ converges in ℓ^1 by the uniform polynomial decay of Part (ii) and $\mathbf{Q}_{L_n} = \boldsymbol{\Xi}'_{L_n}(\mathbf{M}_{L_n} - \rho_1\mathbf{I}_{L_n})\boldsymbol{\Xi}_{L_n}$ is bounded), $\mathbf{x}^*(\infty)$ is a stationary point of the infinite-dimensional problem with multiplier λ°.

Step 5: Uniqueness of the limit. For $\lambda \in J_{++}(\infty)$, define the smooth ℓ^2-valued map $\mathbf{w}_\infty(\lambda) := \lambda\mathbf{K}_\infty(\lambda)^{-1}\mathbf{f}_\infty$, the corresponding stationary profile $\mathbf{x}_\infty(\lambda) := \boldsymbol{\gamma}_{MSE} - \boldsymbol{\Xi}_\infty^{-1}\boldsymbol{\Delta}_\infty\mathbf{w}_\infty(\lambda)$, and the infinite-dimensional secular function $g_\infty(\lambda) := \langle \mathbf{x}_\infty(\lambda), \mathbf{Q}_\infty\mathbf{x}_\infty(\lambda)\rangle$. By Step 4, any subsequential limit $\lambda^\circ \in J_{++}(\infty)$ satisfies $g_\infty(\lambda^\circ) = 0$.

(a) Derivative of g_∞. Differentiating the identity $\mathbf{K}_\infty(\lambda)\mathbf{w}_\infty(\lambda) = (\mathbf{I} + \lambda\mathbf{S}_\infty)\mathbf{w}_\infty(\lambda) = \lambda\mathbf{f}_\infty$ with respect to $\lambda \in J_{++}(\infty)$ yields

$$\mathbf{K}_\infty(\lambda)\dot{\mathbf{w}}_\infty(\lambda) = \mathbf{f}_\infty - \mathbf{S}_\infty\mathbf{w}_\infty(\lambda). \tag{53}$$

Using the definitions $\mathbf{f}_\infty = \boldsymbol{\Delta}'_\infty(\mathbf{M}_\infty - \rho_1\mathbf{I})\boldsymbol{\Xi}_\infty\boldsymbol{\gamma}_{MSE}$, $\mathbf{S}_\infty = \boldsymbol{\Delta}'_\infty(\mathbf{M}_\infty - \rho_1\mathbf{I})\boldsymbol{\Delta}_\infty$, and $\boldsymbol{\Xi}_\infty\mathbf{x}_\infty(\lambda) = \boldsymbol{\Xi}_\infty\boldsymbol{\gamma}_{MSE} - \boldsymbol{\Delta}_\infty\mathbf{w}_\infty(\lambda)$, the right-hand side of (53) factors as

$$\mathbf{f}_\infty - \mathbf{S}_\infty\mathbf{w}_\infty(\lambda) = \boldsymbol{\Delta}'_\infty(\mathbf{M}_\infty - \rho_1\mathbf{I})\boldsymbol{\Xi}_\infty\mathbf{x}_\infty(\lambda) = \boldsymbol{\Delta}'_\infty(\boldsymbol{\Xi}'_\infty)^{-1}\mathbf{Q}_\infty\mathbf{x}_\infty(\lambda). \tag{54}$$

Since $\dot{\mathbf{x}}_\infty(\lambda) = -\boldsymbol{\Xi}_\infty^{-1}\boldsymbol{\Delta}_\infty\dot{\mathbf{w}}_\infty(\lambda)$, differentiating $g_\infty(\lambda)$ gives

$$\begin{aligned}\dot{g}_\infty(\lambda) &= 2\langle \dot{\mathbf{x}}_\infty(\lambda), \mathbf{Q}_\infty\mathbf{x}_\infty(\lambda)\rangle = -2\langle \dot{\mathbf{w}}_\infty(\lambda), \boldsymbol{\Delta}'_\infty(\boldsymbol{\Xi}'_\infty)^{-1}\mathbf{Q}_\infty\mathbf{x}_\infty(\lambda)\rangle \\ &= -2\langle \dot{\mathbf{w}}_\infty(\lambda), \mathbf{K}_\infty(\lambda)\dot{\mathbf{w}}_\infty(\lambda)\rangle \le 0,\end{aligned} \tag{55}$$

with equality if and only if $\dot{\mathbf{w}}_\infty(\lambda) = \mathbf{0}$, because $\mathbf{K}_\infty(\lambda) \succeq c_\delta\mathbf{I} \succ \mathbf{0}$ on $J_{++}(\infty)$.

(b) Non-degeneracy at roots. We show that $\dot{\mathbf{w}}_\infty(\lambda) \neq \mathbf{0}$ for every $\lambda \in J_{++}(\infty)$. By the standing non-degeneracy hypothesis ($\rho_1 \neq \rho_{MSE}$), $g_\infty(0) = \langle \boldsymbol{\gamma}_{MSE}, \mathbf{Q}_\infty \boldsymbol{\gamma}_{MSE} \rangle \neq 0$. Note first that if $\mathbf{f}_\infty = \mathbf{0}$, then $\mathbf{w}_\infty(\lambda) \equiv \mathbf{0}$ and $\mathbf{x}_\infty(\lambda) \equiv \boldsymbol{\gamma}_{MSE}$, which would imply $g_\infty(\lambda^\circ) = g_\infty(0) \neq 0$, contradicting $g_\infty(\lambda^\circ) = 0$; hence $\mathbf{f}_\infty \neq \mathbf{0}$. Now suppose $\dot{\mathbf{w}}_\infty(\lambda) = \mathbf{0}$ for some $\lambda \in J_{++}(\infty)$. By (53), $\mathbf{S}_\infty \mathbf{w}_\infty(\lambda) = \mathbf{f}_\infty$, and substituting this into $(\mathbf{I} + \lambda \mathbf{S}_\infty)\mathbf{w}_\infty(\lambda) = \lambda \mathbf{f}_\infty$ yields $\mathbf{w}_\infty(\lambda) = \mathbf{0}$, which in turn forces $\mathbf{f}_\infty = \mathbf{S}_\infty \mathbf{w}_\infty(\lambda) = \mathbf{0}$, a contradiction.

Consequently, $\dot{g}_\infty(\lambda) < 0$ for all $\lambda \in J_{++}(\infty)$, so g_∞ is strictly decreasing on $J_{++}(\infty)$ and possesses a unique root $\lambda^*(\infty) := \lambda^\circ \in J_{++}(\infty)$. Since every convergent subsequence of $\lambda^*(L)$ must converge to $\lambda^*(\infty)$, the full sequence converges:

$$\lim_{L \to \infty} \lambda^*(L) = \lambda^*(\infty) \in J_{++}(\infty).$$

The asymptotic I-SSA formulated in assertion iii) follows straightforwardly from the above derivations.

□

Remarks: Two structural devices underpin the proof of Theorem 2:

1. *Rank-one boundary completion in Part (i):* Appending the row $-\mathbf{e}_L'$ to $\boldsymbol{\Delta}_L$ to form $\widetilde{\boldsymbol{\Delta}}_L$ (Step 2) completes the rank-one boundary deficit in $\boldsymbol{\Delta}_L' \boldsymbol{\Delta}_L$ to yield the exact Dirichlet Laplacian $\mathbf{L}_D = 2(\mathbf{I}_L - \mathbf{M}_L)$, while simultaneously embedding $\mathbf{K}_L$ as the leading $L \times L$ principal submatrix of $\widetilde{\mathbf{K}}_L$. This allows one to bypass delicate rank-one perturbation and secular resolvent analyses, yielding exact finite-sample two-sided eigenvalue bounds directly via classic Fourier spectral decomposition (despite permanent memory) and Cauchy's Interlacing Theorem. Inverting the ordering in the orthogonally similar $\mathbf{O} := \mathbf{JSJ} = \boldsymbol{\Delta}(\mathbf{M} - \rho_1 \mathbf{I})\boldsymbol{\Delta}'$ is crucial for deriving the exact eigenvalues of the embedding operator via Sylvester's determinant identity, noting that the commuted $\widetilde{\boldsymbol{\Delta}}\widetilde{\boldsymbol{\Delta}}' \left(\neq \widetilde{\boldsymbol{\Delta}}'\widetilde{\boldsymbol{\Delta}} = \mathbf{L}_D = 2\mathbf{I} - \mathbf{M} \right)$ no longer commutes with $\mathbf{M} - \rho_\mathbf{1}\mathbf{I}$ from which (38) is derived..

2. *Deviation formulation and asymptotic vanishing of the cointegration multiplier in Parts (ii)–(iii):* Recasting the system in terms of the deviation from the unconstrained MSE benchmark, $\mathbf{d}_L := \mathbf{x}_L^* - \boldsymbol{\gamma}_{MSE}$, and preconditioning via the congruence $\mathbf{H}_L^{-1} = \boldsymbol{\Xi}_L^{-1} \boldsymbol{\Delta}_L \mathbf{K}_L^{-1} \boldsymbol{\Delta}_L' (\boldsymbol{\Xi}_L')^{-1}$ achieves two goals simultaneously. First, it replaces the ill-conditioned cumulative-sum Hessian $\mathbf{H}_L$ with the uniformly well-conditioned pentadiagonal matrix $\mathbf{K}_L$. Second, because $\boldsymbol{\Delta}_L' \mathbf{1} = \mathbf{e}_L$, the global level/cointegration constraint $\mathbf{1}' \mathbf{x}_L^* = \Gamma(0)$ is localized to the right boundary via $\boldsymbol{\zeta}_L = (\boldsymbol{\Xi}_L')^{-1} \mathbf{e}_L$. Convolving the forward-decaying forcing $\mathbf{K}_L^{-1} \mathbf{f}_L$ with the backward-decaying boundary vector $\boldsymbol{\zeta}_L$ forces the Lagrange multiplier μ_L to vanish at rate $O(L^{-s})$, ensuring that finite-sample boundary perturbations disappear asymptotically without distorting the decay of the filter weights.

Corollary 2

Proof. Theorem 2 ensures that MA approximations are valid, up to negligible deviations. Therefore, the objective function and the smoothness constraint admit their intended interpretations: the MSE (on levels) and the first-order ACF (in stationary differences), respectively.

Consider $\mathrm{MSE}(\rho_1) := J(\lambda^*(\rho_1))$ as a function of ρ_1, with $|\rho_1| < \rho_{\max}(L)$. Eq.(15) shows that MSE is strictly *increasing* on the right branch $\rho_1 > \rho_{MSE}$. By Assertion 4 (global minimum), no predictor can lie below the curve $J(\lambda^*(\rho_1))$, so the corresponding space is empty. We conclude that, for given MSE, I-SSA maximizes ρ_1 on the right branch. Since $1/ht_1 = \arccos(\rho_1)/\pi$ is strictly monotonically decreasing, I-SSA thus minimizes the TP rate $1/ht_1$ for given MSE, assuming a vanishing mean in first differences (otherwise it minimizes the crossover rate about the linear trend). A similar argument applies to the left branch $\rho_1 < \rho_{MSE}$, where the strictly *decreasing* MSE, together with the global minimum, implies that I-SSA maximizes the TP rate. □

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